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A Logico-Philosophical Analysis of Modal and Deictic Phenomena in Natural Language

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Abstract

The philosophy of language, along with formal-logical methods of analytic philosophy and formal semantics, is a relatively young field of inquiry in linguistics, despite being deeply rooted in philosophy since ancient times. The aim of this thesis is to identify and provide, using philosophical argument and logical formulations respectively, the ontological status of modality and indexes as seen in natural language use and their implied truth-conditions. The design follows a descriptive-analytical pattern, with a methodology of tackling each linguistic category and sub-category individually – starting by presenting their best arguments to-date, their detailed critiques and shortcomings, my own approach to the issues, followed by the resulting formal representations and their emergent constraints on meaning. The outcome of this analysis is a distinctly formulated schema of the tackled linguistic phenomena and their implications on their undiluted logical forms prior to the introduction of any context, with the implications of what this signifies for human cognition and the ontology of language, opening the path to further research and establishing the key role of semantics within such inquiries.

Key terms: deixis, formal semantics, logic, modality, ontology

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List of Abbreviations

LS: Logico-Structural system

CTP: Causal-Temporal Projection system

PRO/RET: PROACTIVE/RETROACTIVE

$v(x)$: A valuation function assigning truth-conditions to the expression x

W_x : A specified world

ϕ/ψ : A specified state in the world

$\forall x$: Universal quantifier; “for all x ”

$\exists x$: Existential quantifier; “for some x ”

$\Diamond$: Possibility operator

$\Box$: Necessity operator

$\models$: Semantic entailment

$x := y$: Definition; x is defined as y

WE: Weak essentialism

SE: Strong essentialism

D_E : A domain that allows object-consciousness interaction

OP_E : The emotive potential of an object’s property

$\Delta\delta(x,y)$: A relation along a stretch of time

DRT: Discourse Representation Theory

MSDRT: Mental State Discourse Representation Theory

ER: Entity representation

MSD: Mental state description

dox: Doxastic; related to a subject’s belief state

hn: Human necessity

Phys: Physical

Prox: Proximal

t_S : Subject's anchor time

t_E : Event time

t_R : Reference time

C_x : Causal link(s) connecting events across points/intervals in time

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Chapter 1

Introduction

1.1. Statement of the Problem

The persistent attempt at an absolute, objective analysis of linguistic phenomena has been an unstable endeavor in antiquity and modernity likewise. A primary cause of this instability has been the pervasively elusive nature of semantics in comparison to its partner fields in linguistics. “No area of linguistic theory is more veiled in obscurity and confusion, and it may be that fundamentally new ideas and insights will be needed for substantial progress to be made in bringing some order to this domain.” (Chomsky, 1972, pp. 75–76). Indeed, the formal logics heretofore extant have proven shaky and inadequate at assessing more complex linguistic phenomena in complete consideration to every variable or epistemological gap, and have focused more on simple, declarative sentences with only sparing attention or incomplete analyses to transformations of tense, attitudes, syntactic constraints, presupposition behavior, indexes, and modals. The present thesis tackles the latter two transformations, as they have remained elusive to formal systems of definition, representation, and even philosophical categorization. The basic assumptions of elementary formal logic in semantic analysis are delineated in the literature review, and serves as a basis upon which I present existing theories and form new ones.

1.2. Purpose of the Study

As for my primary objective, this thesis tackles the ontological status of modality and indexicality found in language independent of any cognitive or pragmatic constraints, including how to proceed in laying forth their entailing truth-conditions and formal representations per given type. The veracity of the theories of various formal semanticists,

linguists, and philosophers are critiqued according to my constraints and commitments, followed up by my own re-formulations of the addressed phenomena. The contrast of the semantic representations laid forth in this thesis with the ones in the tackled papers is that of independence from unstable variables such as pragmatic and cognitive elements, which are common in most formal semantic analyses. What is aimed towards in eventuality is a retrospective account of the logical structure of the world using the structure of linguistic meaning as a medium for understanding it. While the scope of this thesis extends only towards modality and deixis as used in language, the same analytic approach may be applied to other linguistic categories as well.

1.3. Significance of the Study

This study undertakes an assessment of a newly emergent domain in theoretical linguistics, one that has often been neglected in modern linguistic studies up to until the last few decades. Through a successful formal semantic analysis and a rigorous philosophical support, the triangular connection between the mind, language, and the world can be made clearer. Specifically, formal semantics generally does away with pragmatic constraints, but only a few do away with cognitive constraints as well. Through the approach used in this thesis, the exact transitional phase between the objective state of the world and the human mind's reception of this objectivity, before the introduction of pragmatic and cognitive constraints, can be traced to its origin-most point and laid forth representationally. By so doing, we will be able to apprehend, to the extent that the human being might be capable, the most primordial and undiluted form of meaning. To do so, the stability of the contemporary theories in formal semantics pertaining to the two linguistic categories tackled here must be examined, and as needed, improved upon by eliminating the above-mentioned unstable variables. This is how a more objective account of the world via language might be reached

and my purpose achieved: A more rigorous formal-logical analysis of language that represents more accurate formulae of world-states.

1.4. Overview of the Philosophy of Language & Formalism

Before presenting the consequent outline or foundational assumptions, it would be apt to provide a general definition of formal semantics and the philosophy of language, including their scopes as fields and what they aim to accomplish as they have thus far been done. It must first be acknowledged that formal semantics has its roots originally in the philosophy of language (Portner & Partee, 2008). The philosophy of language can be traced to classic texts of philosophy, some of which will be mentioned in the literature below. As a field, Lycan (2018) defines the philosophy of language as a domain that takes various forms, which includes investigations via the philosophical analysis of linguistic phenomena such as referencing, entailments, presuppositions, implicature, truth-conditions, necessity and possibility, intensions and extensions, and speech acts. Indeed, the area itself may be approached either semantically or pragmatically, with examples being Frege and Grice respectively. The essential objective of both approaches is that they aim to understand the dimension of language that exists beyond the level of empirical linguistic analysis, observation, or quantification. Miller (2007) acknowledges that the great “leap” from the previous system of Aristotelian logic to the more complex, modern logical systems can be greatly attributed to Frege and his refinement of the language of logic in his *Begriffsschrift*, which is expounded upon later. As far as the distinction between semantics and formal semantics goes, Chierchia (2006) explains that while semantics as a field is concerned with the question of what interpretation should be assigned to various linguistic symbols, formal semantics does so in a rigid, mathematical methodology. Portner and Partee (2008) point out a significant difference between other approaches to meaning analysis and that of formal semantics. As seen in many cognitively-based views including the likes of Jackendoff, Fodor,

Chomsky, and Kamp in their general works, the primary underlying argument for them is that meaning is representational, hence positioning meaning as one of the functions or modular processes of the mind (Jackendoff, 1988) regardless of whether or not the theorist advocates for a universal, objective approach to language. On the flipside, Portner and Partee posit that formal semantics utilizes a model-theoretical approach, which comprises the artificial language of different systems of logic such as propositional, predicate, and modal logics, hence making it a formal system of deliberation and assessment. Novaes (2012) defines the concept of formality in linguistics and philosophy as an approach to analysis that maintains neutrality, isolates itself from speaker intentions, or in some interpretations, a step away from semantics and into more rigid systems like syntax. A key property that she notes, however, is that formal languages possess the merit of being computable, and hence being accurate in their representations and interpretations, regardless of which definition of “formal” is adopted. A formal approach is generally associated with objectivist views of language, as similarly applies to this study, for the models of language that are presented are often said to reflect factual, structural truths and falsities or other such entities as properties and ontological states rather than cognitive processes, which would fall on shakier grounds of formality. The degree of their formality, however, is the issue under constant scrutiny.

1.5. Research Questions

As a consequence of using formal mechanisms in order to reach world-based truths, a set of questions about the stability and veracity of these systems arise:

1. To what extent can we ascertain the definitive truth-values of modal and deictic propositions using purely logical systems of representation?
2. What are the necessary ontological commitments that must be addressed before attempting formal representations of modality and deixis?

3. How do the theories of other linguists and philosophers hold up against the rigid objective commitments of formal semantics?

1.6. Definition of Terms

The following list contains key terms that are relevant to one's understanding of this thesis and the field under which it belongs: Formal Semantics.

Content/Character: The content of a proposition is its meaning in a particular context as it is assigned one via its character in that specific context. These are Kaplan's original terms that I criticize in the body. (Kaplan, 1979)

Deixis: A linguistic category of terms that contains spatial, temporal, and personal indexes whose interpretation depends on context. (Levinson, 2006)

Event Time, Reference Time, Speech Time: Semantic parameters that assess the temporal location of an event, the interval in which it belongs, and the time of utterance respectively. These are Reichenbach's original terms that I formalize further, adapt, or omit in the case of speech time. (Reichenbach, 2005)

Event-time: An expression I take to refer to a particular event occurring at a specific time within an innate domain of self-definition.

Extension: The truth-value of a proposition in a given world. (Asher, 1985)

Formal Semantics: The application of model-theoretic logic to semantics in order to accurately provide truth-conditions for linguistic expressions as they relate to formal structures. (Portner & Partee, 2008)

Intension: A function that maps an index or interpretation to all possible worlds, yielding a universal truth-condition. (Carnap, 1988)

LS System: A formal structure I propose to be the link between language and the world; an ontology of logical structure in the world out of which meaning is derived, and the subject-matter of formal semantics.

Modality: A linguistic category of terms that contains terms of necessity, possibility, and dynamic expressions of obligations and attitudes. (Usmanova, 2021)

Model-theoretic semantics: A framework that analyzes meaning according to a particular domain of individuals, times, possible worlds, or other similar structures. (Portner & Partee, 2008)

Ordering Source: A semantic tool in modal logic that arranges possible worlds according to a criterion of evaluation or preference. (Kratzer, 2012)

Possible worlds: Abstract semantic tools, or philosophical assumptions, that are used to assess the truth-value of a proposition relative to whether they are necessary or contingent across worlds – the ways things could have been. (Girle, 2014)

Semantics: The study of meaning in language via a truth-conditional approach independent of pragmatic constraints. (Chierchia, 2006)

Truth-value/Truth-condition: The conditions under which a sentence would be true. The truth-conditions of a proposition maps to it a particular truth-value of true or false (1 or 0). (Kemp, 1998)

1.7. Structure of the Thesis

This thesis follows a systematic pattern of tackling each linguistic case of modality and deixis separately. The analysis is divided into two major sections, one for each mentioned domain, each divided into sub-sections pertaining to a specific lexical unit of that category in both isolation and some embedded linguistic contexts. Each major section will begin with a

brief delineation of what is to be found conventionally in that category's linguistic repertoire, while each sub-section begins with the presentation of a well-known theorist's research into that specific area, followed up by my scrutiny of their arguments for any logical or philosophical gaps, leading up to a re-tackling of the presented issues into my own more rigid, formal-logical representations. This makes my current methodology of analysis a combination of philosophical argumentation and the mathematical-logical array of formal semantics. This pattern will persist until the final section of the thesis, in which I will illustrate all of my formulated or reformulated formulae into a succinct chart of objectivist semantic identities.

1.8. Assumptions and Implications

It would be fitting to now present the foundational assumptions of this thesis. At its roots, my approach towards semantic analysis presupposes an objectivist, universal view of language as an entity. A skeptical approach towards such absolutist analyses of meaning is a veritable one. The general consensus in linguistics supports relativistic analyses of linguistic phenomena, as seen in many pragmatic approaches, discourse analysis, typology, and other cross-linguistic studies. It must be stressed that this work does not assume a homogeneous structure of language and is not in contradiction with the heterogeneity of language use. It does assume, however, a Platonic and Cartesian view of language as a pre-contextual, objective entity. While a metaphysical argument for this position is worthy of its own dissertation, a brief argument for it will be given here to alleviate the metaphysical burden of the proceeding theories.

The underlying claim to my arguments, and the primary claim which I aim to only solidify by them, is that there exists a layer of language that persists universally across all linguistic forms, which becomes heavily modified after the introduction of pragmatic and cognitive processes. Moreover, the origin of this universal structure is found in the world

itself, via its own logical and highly rigid structure - that which the mind attempts to interpret, throughout which a multitude of variables are introduced that are impertinent to the world's structure itself. This is hence why I refer to my view of language as Platonic and Cartesian, as it understands language as a split system of subject-meaning and object-meaning, the latter of which can only be reached through rigorous semantics. What my formal semantics aims to establish through tackling various linguistic phenomena is this very notion: It is only through the purest semantic distillation that meaning, as it reflects the objective world, can be identified. Thus is the primary objective of any consequent analysis.

Chapter 2

Literature Review

2.1. Historical Accounts on Meaning

It would be prudent to briefly overview how formal semantics came to be. The great inquiry itself has been extant since the classical era of philosophical discussions aimed towards language, with the most notable example being Plato's *Cratylus*, which can be seen as an original, focal investigation into the now-informal question "How are names assigned to their references?", serving as a precursor to the contemporary issue of reference in semantics. The text is centered around an argument between Hermogenes and Cratylus, whose philosophical positions reflect the modern "conventionalist" and "naturalist" linguistic ideologies respectively (Sedley, 2003). Linguistic naturalism is strongly rooted in metaphysics, and is often attributed to ancient writers. De Jonge (2019) notes how Ancient Greek and Roman authors frequently viewed language as an ontological representation of reality and that the true potential of language is to imitate nature with the greatest accuracy - a position adjacent to the presuppositions of this thesis. Cratylus is observed to argue that there exist accurate names and inaccurate ones, such as random sounds; furthermore, the aforementioned accurate names are so in virtue of them reflecting the very essence of the referred object. This extreme view is made even more intangible when he claims, in accordance to Heraclitean doctrine, that names are the only medium through which one can identify and grasp the truth of objects as found in their perpetual, eternal flow (Allan, 1954). Conventionalism, in contrast, is a more prevalent view in contemporary linguistics; its core thesis states that the linguistic processes such as naming, along with all linguistic "rules", are the result of a longitudinal, diachronic repetition of those very rules by their speakers. Through time, these conventional rules embed themselves into language and adopt the

identity of prescriptive linguistic rules. Gionnani (2017) sums up the phenomenon of a linguistic convention effectively: “The existence of certain linguistic habits relevant to the use of a sentence S is a necessary and sufficient condition for the necessity of the proposition meant by S.” (p. 129). In other words, in a conventionalist ideology of language, a proposition means what it does in virtue of the linguistic habits relative to its use. Such ontological issues deserve their own theses, and are to be recognized as still relevant if scrutinized with the appropriate philosophical depth.

The proposition that language reflects the world itself, either fully or partially, can be seen echoed in formal semantics. Chomsky (1995) says as much by specifying that words in natural language can be assumed to pick out either “(1) things in the world, or (2) things in some kind of mental model, discourse representation, and the like.” (p. 24) and adds to the former that “we could then construct truth theories, and develop an account of communication through shared entities...” (p. 24). As stated before, this is the view followed by this thesis. Hence, to do this, philosophers and linguists have often used the language underlying mathematics – formal logic – to account for abstract meanings beyond the observable dimension, and the inklings of this has evolved in history in a parallel manner. Van Fraassen and Fraassen (1971) mention how even the earliest instances of describing systems of logic involved an analysis of language in parallel to it, as the two cannot be isolated. A well-known historical example they offer is William of Ockham’s *Summa Logicae*, in which he divided his analysis of logic into the study of terms, propositions, and argumentation. In order to properly address the rules of argumentation, a linguistic analysis of the various types of terms, which corresponds to grammatical categories, and how they interact with one another, is essential. Consequently, the same process is applied on combinations of terms into well-formed formulae called propositions, and once again, how they interact when juxtaposed. Van Fraassen and Fraassen also note that semantic analyses of

such sorts always bring about the use of a metalanguage; however, this metalanguage may be natural or artificial. For the approaches mentioned so far and as it is related to this thesis, semantic analysis often incorporates the artificial language of logic to ascertain clarity and compositionality. Nonetheless, there are linguists who advocate the position that semantics can be complete with natural language analyses as it is, for the concern that artificial languages might sully natural meaning (Goddard, 2008).

Another well-studied account of the meaning-functions of language is given by John Locke. In his analysis of words and signification, he argues that the only function of language is to make manifest the ideas within the subject's mind, holding that there is no referential property to linguistic units or a resulting connection to truth. Essentially, for Locke, words are nothing more than a direct and immediate representation of the speaker's ideas, through which the world is accessed (Kretzmann, 1968). This view is quite mentalistic, especially as it assumes the link between language and the world has a mediator, and could be argued to be an early form of a cognitivist view of language - one opposed to the central claim of this thesis. Indeed, Locke generated a fair amount of controversy with this argument. As Ott (2003) explains, Locke has been criticized for a variety of shortcomings, such as not accounting for abstract, universal terms and how we may share such concepts in mutual understanding. Moreover, if ideas are an essential mediator between the world and the language, this by no means is a sufficient indicator that words signify those very ideas. Ashworth (1984) points out another inconsistency in Locke's argument, which is his treatment of linguistic units such as particles and function words, like "not". Locke claims that such units do not, in fact, reflect individual ideas, but rather represent an action or attitude of the mind towards an idea, similar to the modern concept of predicates. This is a clear exception to his previous thesis. Lastly, Ashworth (1984) provides an in-depth discussion of Locke's conception of language. He points out the fact that through Locke's

emphasis of ideas, he could be considered a precursor to Wittgenstein's private language theory, which states that language operates purely cognitively and pragmatically on an individual basis, rendering an objective analysis of language moot. Locke also commended mathematicians for casting names aside and analyzing ideas in their barest form, a highly relevant statement in the context of formal semantics, for this proposition harmonizes with the methodology of formal semantics, but the underlying ontology between it and Locke differs greatly. A last point that Ashworth sheds light on ties in to my proceeding overview of Grice's distinction between natural and non-natural meaning. Locke's commitment to ideas over matter is once again acting as a precursor, but to Grice's conception of non-natural meaning, which is an important concept for the content of this thesis, expounded upon soon.

2.2. Ontology and Commitments

A point that ought to be elaborated on before Grice's claims, however, is the idea of ontology and ontological commitments. Schroeder (2011) stresses the fact that formal semantic must be informed by philosophical investigation in order to clarify what the given symbols and their interpretations signify of the world. Thereby, when it comes to world, the philosophical field of ontology comes into play. Smith (2012) offers a clear explanation of ontology by stating that ontology is the study of what is, along with the potential and possible structures of reality and the relationships, characteristics, and all general happenings within them – "Ontology seeks to provide a definitive and exhaustive classification of entities in all spheres of being." (p. 47). Consequently, an ontological commitment, he adds, is a view largely championed by Quine in his discussions on the scientific enterprise that posits any scientists to be committed to the existence of certain entities before the scientific investigation begins; this might take the form of elementary logical axioms or the existences of entities such as electrons, or motion. Most essentially, Quine (2013) himself holds these commitments exclusively take the form of formal logical language, as it is the only one

without ambiguities or other embedded commitments; this can be exemplified in his famous statement “To be is to be the value of a bound variable”, meaning that for any entity to exist in a linguistic or mental formulation, such as “The cat ran down the hall.”, the basic assumption of “There is a cat”, or $\exists x(x \text{ is a cat})$, must be assumed. This is essential to this thesis, as I do seek to “reach” the ontological level of common linguistic functions; furthermore, my ontological commitment goes beyond this level to what can be best exemplified through Grice.

2.3. Grice’s Language Ontology

In light of my ontological commitment briefly mentioned in the introduction, an overview of Grice’s natural and non-natural meaning is now appropriate. It must be noted that this distinction and its bearings on the ontology of language ought to shoulder its own dissertation, and so my review in this section will be brief and provide only the gist of Grice’s contribution to my analyses. Grice (1957), before his work in pragmatics, thought it fit to distinguish between two metaphysics of linguistic meaning – natural and non-natural (Wharton, 2003). Natural meaning can be understood as metalinguistic meaning, or meaning independent of language itself. This can be mainly illustrated by the natural causality found in the matter-based world and the resulting relations; for instance, a cloudy sky means it is going to rain, or a moldy room means it has been unkept. Such relations are language-independent, and point towards meaning that pre-exists language; this is an essential bearing on my formal semantic analyses, as it is this form of meaning that is being decoded – one originating from the world and the commitment I adhere to. Non-natural meaning, on the other hand, is the meaning analyzed much more commonly by linguists, as it is the meaning that is formed post the psycholinguistic processes of encoding and hence becomes identifiable as intended meaning (Davis, 1998).

2.4. Formal Semantics

Moving forward in time from the philosophical to the formal, the movement of semantics from the relatively vague, formless questions of the ancients to more modern, scientific analyses that more reliably focus on symbolic methods and meaning as found “out there” can be largely attributed to the 20th century mathematician and philosopher, Gottlob Frege, who set down the essential fundamentals for the inquiries in Formal Semantics, albeit he did not name the field itself, and laid down the path for Russell, Carnap, and eventually Montague, to follow (Partee, 2011). While his developments in *Begriffsschrift* are more relevant to pure mathematicians, his idea of Sense and Reference in *Sinn und Bedeutung* is critical to those who intend to build on his framework in semantics. Frege (1892/1997) notably distinguishes between the sense and reference of linguistic expressions, with the former denoting how words carry with them different mental and logical representations and the latter denoting the literal entity that is referred to in the world. This places Frege on precarious grounds, as while his notion of reference is quite typical, the idea of sense remains theoretically shaky. Frege seems to propose an abstract “logical realm” that exists alongside empirical reality, and this has some parallels to this study, as I also claim that language is rooted in logic. This logical space represents a collection of functions and predicates that reside over all things – a view developed first in mathematics by Frege himself, which extends to language. This is a purely recursive theory, seeing as how even atomic statements such as “Dog” must correspond to an abstract predicate $\Phi(\text{dog})$ that necessitates the quality of “being a dog”, and that quality will have its own predicate, and so on (Tichý, 1988). This theory is not without criticisms, especially in regards to its infinite regress, yet my own view is similar in attempting to reach the logical realm that fundamentally exists beneath the empirical world. An essential difference between Frege and the semantics I adopt, however, is that Frege views logical axioms as the terminal point, while I take primary ontological

variables, such as causality and time, to be terminal points, only coupled with logical axioms. Lastly, Frege still acknowledges mental constraints on meaning, and this is rejected by this thesis, as I hold that mental variables are only introduced to meaning after the fact that logical world-structures already exist. After Frege's foundation, it was figures such Russell and Carnap who made further leaps in the formal analysis of language, leading to and solidifying Richard Montague's distinction between extensional and intensional approaches to semantic analyses, which are key divisions in the field to this day. Dowty, Wall, and Peters (2012) provide a concise representation of Montague's work, and while the detailed properties of Montague Semantics aren't in the scope of this literature review, it is worth noting the ideas of intensions, extensions, and lambda entities, as they are the root of his work and the majority of consequent semanticists, originally championed by Carnap. The most explicit form of semantics will be addressed first: Extensional semantics. Asher (1985) defines extensional semantics as an attempt of laying forth the adequately descriptive truth conditions of a proposition in reference to a given world, often the local one. This means that an extensional analysis is done by rooting itself in the current, empirical world, and treating its variables as stable entities and members of sets (Partee, Ter Meulen, & Wall, 2012). Dowty et al. define intensions as the explicative function that determines the truth or falsity of a proposition at any given index (time and world-state) in reference to its denotation (extension). In other words, they are formulae that consider a particular reference in the context of any possible state-of-affairs and return a truth-value. Essentially, an intensional semantics will determine when a proposition is true after any relevant variable that might affect it are considered; it does not simply check for truth in the current world as extensional semantics, but rather under what conditions it may or may not be true universally. This implies that intensional semantics is more equipped to handle issues that go beyond static world states and are subject to uncertainty and change (Asher, 1985). This endeavor also

often leads to the use of lambda abstractions, first incorporated by Montague. To understand the lambda abstraction, it must be considered that a common assumption in formal semantics is that the predicates of linguistic entities originate from sets – sets that contain the objects belonging to or having particular properties. Firstly, in a simple declarative as “Robert walks”, Heim and Kratzer (1998) explain that “Robert” functions as an entity, which is designated as “type e”, while “walks” functions as a predicate of type $\langle e, t \rangle$, meaning that it takes an entity of type e and returns t, a truth value. This can be formalized as $\text{walks}(\text{Robert})$, with “walks” as a function that takes the argument “Robert” to give us a truth-value. Lambda calculus comes in by representing the function as an open function $\lambda x.\text{walk}(x)$, which designates the function that returns 1 (true) if x walks, and 0 (false) otherwise. More formally, this may also be represented as $[\lambda x: x \in D. x \text{ walks}]$; D stands for the domain of definition in which x is defined, which is commonly a world W and time T with more or less specification given the complexity of the analysis. Then the formula expresses that every x in D is true if and only if x walks, and false if not. Were we to factor in an entity e into this formula, it would appear as $[\lambda x: x \in D. x \text{ walks}](\text{Robert})$; this applies the entity “Robert” to every instance of x in the function of type $\langle e, t \rangle$, which, if we simplify by removing the domain definition, returns us to $\text{walks}(\text{Robert})$. After this, it is checked empirically whether Robert is indeed part of the set of walking entities in the given domain. Of course, this is only the most rudimentary of analyses as it only contains one entity and one predicate; as we will see in my subsequent analyses, the issue becomes much more convoluted with the introduction of modal, spatial, and temporal constraints.

Extensions are related to another famous issue debated historically, and that is Russell’s theory of definite descriptions and his contemporaries’ responses to it. This issue is relevant to us as it concerns the ontology (or lack thereof) of linguistic expressions, apparent in the most rudimentary manner within simple, declarative sentences, which Russell

addressed as definite descriptions. Albrecht (2007) lays forth this foundational issue in semantics by asking whether all expressions in language must necessarily have denotations (extensions) or is it plausible to claim that not all linguistic terms have referents. If so, on what grounds shall one consider the truth or falsity of that term or the expression in which it is found? Parsons (1995), in a philosophically provoking article, illustrates Alexius Meinong's point of view. According to Meinong, all linguistic expressions refer, but they may refer incompletely to entities that lack existence. In other words, in Meinong's ontology, we have entities that exist, and entities that don't exist, but both have their place in the Universal ontology as objects that have Being. This is understandably a view that merits argument, but without doing so, it is simply a perspective on linguistic ontology that one may consider. Unsatisfied with this theory, Frege proposed his famous distinction between sense and reference, explained above, to account for how a linguistic term may contain psychological significance without necessarily denoting an object in the world (Albrecht, 2007). In a notable attempt at avoiding the psychological dependence, and hence precarious nature, of Frege's sense and reference theory, Russell (1905), in his influential paper, On Denoting, argued that sentences in the form of "The x is y" do not, in fact, directly refer to an entity [x] or presuppose it, but rather subtly affirms the existence of x alongside making a claim about it. For instance, in an expression analogous to "The apple is round", we have not just an act of predication (ex. $\text{round}(\text{apple})$), but also an implicit truth-claim about the apple itself, implying, given the determiner "the", that "there exists at least one apple", thus driving it closer to a logical entailment rather than a presupposition from a receiver. Russell hence interpreted such a sentence in his trademark manner: "there exists at least one apple (x), and that apple is round. If another entity exists that is round, it is identical to the apple (x)", or more formally, $\exists x(F(x) \wedge \forall y(F(y) \rightarrow y = x) \wedge G(x))$ (Irvine, 2024), thus avoiding issues with non-existent referents by simply holding them as false in virtue of the former half of the

formula, “there exists an x”, being false. Of course, completely adopting this view leads to a metaphysical issue we often pass by in conversation, and that is how we ontologically commit ourselves to quantificational existences that are not empirically sound. As was just seen, Russell’s approach implies that only one x can be y if a definite determiner is used; however, without Russell’s theory, Frege’s and Meinong’s concern will need to be readdressed: How does an expression with a non-existent referent have meaning? A final noteworthy comment on this issue was Strawson’s response to Russell’s theory. He responded by arguing against the axiom of logic that maintains that any proposition must be either true or false, and claims that a sentence with a non-existent referent is neither true nor false, but simply fails to attain a truth-value since, instead of referring to nothing as seen in Meinong’s argument, it simply fails to refer. It seems that either empirical validity or logical rigor are sacrificed in favor of an answer to the ontological commitment level of linguistic expressions in the literature. This section can be concluded by acknowledging the ontological maze that language presents us and one of the foundational questions in this thesis: To what extent do linguistic expressions reflect the structure of the world? As my argument is “To a great extent”, a primary lesson may be taken away from Russell’s and his contemporaries’ discussions that language, and particularly linguistic meaning, cannot be analyzed without committing ourselves to the existences of entities in the world, whether empirical (extensional) or otherwise.

Rightfully so, it is not just empirical discussions that comprise semantic analyses, as a great number of linguistic expressions function beyond simply denoting or marking some form of significance or predication; language also functions in the medium of space, time, and uncertainty. This is where a movement from extensions to intensions is manifested. Some of the well-known philosophers to introduce us to and provide us an intensional semantics are Carnap, Kripke, Lewis, and Davidson. Rudolf Carnap laid down what is considered one of

the original definitions of intension. Building on what came before him, he emphasized that analyzing linguistic meaning must go beyond the empirical level, and this is where my own argument takes its root. He argued that any linguistic entity, predicate, or sentence, possesses not only an extension in the actual world, but also a corresponding concept attached to it; that is, an intension. This may appear analogous to Fregean sense, but Carnapian intensions rid themselves of the psychological burden found in the Fregean sense, and add to it that intensions make use of “models” that take into account the state of the Universe across possible worlds and times, a feature lacking in extensional semantics. They are essentially, as the classic definition goes, functions from possible worlds to extensions. Furthermore, he stresses how extensional semantics requires empirical investigation at its end, while intensional semantics is concerned only with the semantic structure of the given language and the concepts that are conferred by them (Carnap, 1988). This makes an intensional analysis of linguistic meaning a lot more philosophically inclined and burdened, as the truth or falsity of a proposition, entity, predicate, or event, has to be explicated in rigorous detail due to the lack of contextual, worldly constraints.

Kripke contributed to this view immensely through what opposed the descriptive theories of the time. He already hinted early on that semantics cannot be complete by purely rooting ourselves in the “current” world:

... we must suppose that in each world a given n -adic predicate letter determines a certain set of ordered n -tuples, its extension in that world. Consider, for example, the case of a monadic predicate letter $P(x)$. We would like to say that, in the world H , the predicate $P(x)$ is true of some individuals in $\psi(H)$ and false of others; formally, we would say that, relative to certain assignments of elements of $\psi(H)$ to x , $\phi(P(x), H) = T$ and relative to others $\phi(P(x), H) = F$. The set of all individuals of which P is called the *extension* of P in H . But there is a problem: should $\phi(P(x), H)$ be given a truth-

value when x is assigned a value in the domain of some *other* world H' and not in the domain of H ? Intuitively, suppose $P(x)$ means 'x is bald' - are we to assign a truth-value to the substitution instance 'Sherlock Holmes is bald'? Holmes does not exist, but in other states of affairs, he would have existed. (Kripke, 1963, p. 65)

What this means to say is that any given world, such as the real world (W_0), axiomatically has a collection of predicates that either apply or don't apply to a collection of entities; this much has always been a given. However, an issue arises, as we saw before, when we attempt to predicate a quality unto a non-existent entity, such as the Sherlock Holmes example. Avoiding the solutions of Meinong, Frege, and Russell, Kripke eventually concludes that entities, predicates, and their truth values ought to be judged according to whether or not they apply in a specified world (with specified conditions), not an arbitrary one or simply the "real" one. Currently, such worlds are specified by domains (D), which comprise world-states (W) and even different representations of time (Datta, 2023). It is precisely these discussions that led to Kripke's introduction of rigid designations and the notions of necessity/possibility in semantics. In *Naming and Necessity*, Kripke (1980) introduces a concept called rigid designation, which serves as a valuable pillar for the arguments of this thesis. He explains that rigid designators are words that necessarily refer to the same extension in all possible worlds, or in any possible state of affairs, and an example of this is names. Names, according to Kripke, always designate a specific individual metaphysically, even if that individual is pragmatically named differently in a possible world, the name X will always refer to the same ontological entity. In order to accept this theory, it is apparent to us how this must be thought of anti-psychologically, as Frege's original intentions, and adopt an ontological perspective. This is formulated best in this passage:

The tendency to demand purely qualitative descriptions of counterfactual situations has many sources. One, perhaps, is the confusion of the epistemological and the

metaphysical, between a prioricity and necessity... Generally, things aren't 'found out' about a counterfactual situation, they are stipulated; ... (Kripke, 1980, pp. 200-201)

In sum, this means that it is a mistake to attach identity to an entity via mental associations with their corresponding names and their associated properties; names and properties are already givens, which remain fixed across all possible worlds in which those entities exist.

This endeavor towards widening the scope of semantics was expedited greatly by David Lewis. Lewis (2013) discusses what it means for semantics to adopt a possible worlds approach in its methodology. He claims that possible worlds are not just a tool used for linguistic analysis, but a metaphysical truth about the Universe. The most jarring evidence to point in this direction are counterfactuals in language, seemingly pointing towards non-existent world-states in their propositions. In a perfectly innocuous sentence like “If Tom had a mansion, he would decorate it.”, Lewis maintains that the proposition at hand is to be interpreted relative to any given worlds in which Tom, in the manner explained by Kripke, exists, one in which someone may own a mansion, and one in which Tom is allowed or has the means to own one; all of these phenomena are, according to Lewis, accessed from the point of view of our current world to other possible worlds that allow the abovementioned scenarios in their world structure. This latter phenomenon he dubs “accessibility”, which stands for the scope of one world in relation to what it may or may not access relative to other worlds and their laws or limitations. To integrate this into semantics, he introduces two essential operators that shall be used in my own analyses in this thesis: Necessity ($\Box$) and Possibility ($\Diamond$). Essentially, the necessity operator, when used within a formula, holds its arguments to be true in all possible worlds, while the possibility operator limits its arguments to specific worlds. We may reiterate this in his words:

Necessity is truth at all accessible worlds, and different sorts of necessity correspond to different accessibility restrictions. A possibility operator, likewise, is an operator that acts like a restricted existential quantifier over worlds. Possibility is truth at some accessible world, and the accessibility restriction imposed depends on the sort of possibility under consideration. (Lewis, 2013, p. 5)

Lewis' liberal use of modal realities in his semantics stands as another pillar for my own framework, as it too holds modal realities to be properties extant in the world beyond what is cognized by the linguistic receiver's mind.

At this point, it would be reasonable to illustrate how other philosophers adapted intensional semantics to account for various linguistic regularities. An example of this would be Donald Davidson's event semantics, which accounts for the linguistic expression of action. Maienborn, von Stechow and Portner (2011) explain how Davidson's fundamental proposition was the actions described by verbs are ontologically present states of time and space, or in other words, ontological "events", instead of cognitively held descriptions or beliefs. Maienborn et al. provide the following formulae to illustrate an example of a Davidsonian description of an event and a stricter, Neo-Davidsonian one respectively:

- a. Jones buttered the toast in the bathroom with the knife at midnight.
- b. $\exists e$ [butter (jones, the toast, e) & in (e, the bathroom) & instr (e, the knife) & at (e, midnight)] (p. 235)

In such a case, an event e is held to exist, of which "Jones buttered the toast", "In the bathroom", "With a knife", and "At midnight" are all a part as modifiers upon the verb. The benefit of this simple inclusion of the event variable e is that adverbs and adverbial phrases may now be accounted for as modifiers upon the events themselves rather than the linguistic verbs, which makes their analysis more independent and stable within a framework, as

originally held by Davidson (Lasnik, 2006). On the other hand, a Neo-Davidsonian account of the same sentence would read as:

$\exists e$ [butter (e) & agent (e, Jones) & patient (e, the toast) & in (e, the bathroom) & instr (e, the knife) & at (e, midnight)] (Maienborn, von Stechow & Portner, 2011, p. 244)

It is noteworthy here how the existences of agents and their themes are held to be separate events as well rather than arguments for the verb. This analysis designates what one might normally assign consider a verb's arguments their own thematic roles, and this allows the consequent analysis to be more detailed, as all predicates are in and of themselves events.

While the notion of what constitutes an event is philosophically blurry, I will be making use of the event semantics championed by Davidson with my own approach, and thus this section is apropos. My formulae will not adopt his exact framework or compositional rules with arguments, yet events do exist within my upcoming framework, as they cleanly comply with my ontological approach.

This tradition of representing language and meaning via the means of model-theoretic symbols and systems has been the consistent methodology within Formal Semantics, and this is underpinned by various rules proposed by different theorists. Most prominently, we may attend to the principle of compositionality, which was originally set down by Frege according to most sources. Pelletier (2016) provides for us the original understanding of this notion as “The meaning of a syntactically complex unit is a function of the meanings of the complex unit's simpler parts and the syntactic way those parts are put together.” (Pelletier, 2016, p. 1). In other words, the meaning of sentence can be derived from the meanings of its linguistic constituents and how those constituents are combined. Pelletier (2017) explains how this definition of compositionality has its roots in a grander philosophical issue concerning atomism and wholism, with semantic compositionality being backgrounded by the former.

Essentially, the ontological assumption of atomism suggests that any perceived “whole” is neither more nor less than the summation of the parts that make it up, with no unique identity of “wholeness” in itself, while the assumption of wholism suggests that a perceived whole may contain properties not found in its individual parts, and that the whole harbors a unique identity by itself. This debate has heavy bearings on the linguistic quest of establishing a syntax-semantic interface, as either assumption will completely reformulate the analyses established in either field. Most contemporary researchers take semantic compositionality, and hence atomism, for granted as the foundational principle upon which all analyses and formulae function (Partee, 1995); moreover, this incentivizes an analysis of syntactic as well as semantic phenomena in tandem to reach a proper understanding of meaning, for a syntactic shift will change semantic interpretations, and a semantic reformulation of meaning will have consequences on how the rules of syntax function to combine said meanings. This is best put by Partee: “Semantics is syntax-driven, syntax is semantically motivated.” (Aloni & Dekker, 2016, p. 11). As such, much of semantic analyses precede syntactic ones in order to be tested in broader linguistic contexts, and thus semanticists ought to ascertain whether their representations can function when placed in natural language sentences of varying syntax. A way in which formal semanticists do this is called Functional Application (FA), which, as we have noticed before, means that all meanings are essentially represented as arguments upon functions in a particular sequence, given formally by Heim and Kratzer in their discussion of a syntax-semantic interface: “If a is a branching node, (β, y) is the set of a 's daughters, and $[\beta]$ is a function whose domain contains $[y]$, then $[a] = [\beta]([y])$.” (Heim & Kratzer, 1998, p. 44). This implies that a syntactically “higher” linguistic item is given meaning according to the predicate-argument functions of its syntactically lower items; for example, in a node such as “likes Mary”, the meanings of the lower nodes “likes” (a predicate) applies to the meaning of the node “Mary” (an argument) to yield “likes(Mary)”, a function that determines the

meaning of the mother node. Thus, we may see how semantics needs syntax in its system to properly account for how linguistic meaning is structured in different word sequences and combinations.

This may appear to proceed without issue, but compositionality appears to break in certain cases when another phenomenon known as semantic projection emerges. Asher and Lascarides (1998) define presupposition as an assumption that emerges from a clause that must be accepted as true in order for the clause in question to be considered true; however, they do note how it is debatable whether this phenomenon can be accounted for semantically, as the assumption arises from context, making it a pragmatic phenomenon. This is one of daunting challenges of formal semantics, as the primary rule of compositionality and functional application seem to break when the presupposition of a sentence fails to emerge and seems to transform into an entailment when embedded in a complex syntactic structure. This is illustrated in the following example by Rothschild:

Consider for instance, sentences (1) vs (2), adapted from van der Sandt (1992); the presupposition triggered by Jack's son (that Jack has a son) is implied by (1), but not by (2).

(1) If baldness is hereditary, then Jack's son is bald.

(2) If Jack has a son, then Jack's son is bald.

The challenge for a formal semantic theory of presuppositions is to capture contextual effects such as these in an adequate manner. (Rothschild, 2011, p. 1)

We may clearly see how the very same clause presupposes “Jack has a son” in one embedding and not in the other according to the clause(s) adjacent to it. This issue has been tackled immensely in the literature, yet it does not directly concern the topic of this thesis,

and such we are only mentioning it as one of the foundational concerns in formal semantics, even if it is not tackled here.

Skepticism towards the use of logical symbols hasn't been absent since its inception. Lewis (1976) made the claim that a study in semantics cannot be undertaken without addressing truth-values and truth-conditions. This statement translates to the argument that semantics solely done through symbolic translations is inadequate and says nothing about the target language. Rather, he argues, semantics must be primarily done by the specification of the correspondence (or other) conditions that make sentences true, and hence, meaningful. As a brief overview of the various conditional approaches to truth-conditions, one may look at the following prominent theories. According to Soames (1984), one of the earlier notions of truth that deviated from the standard empirical understanding is the coherence theory of truth. It is explained that in the framework of this theory, a proposition is considered true if it corresponds with a given set of facts that are held to be true in a given model, which is meant to stand for a collective mental model. In simpler terms, a statement is true if it is commonly believed to be true. This makes the coherence theory one that is grounded in the stability of models that are ideally, psychologically, and sociologically structured. On the other hand, the pragmatic theory of truth holds that a statement is true or false in accordance to how impactful on the world or individual its truth or falsity is, as put by William James:

Pragmatism, on the other hand, asks its usual question. "Grant an idea or belief to be true," it says, "what concrete difference will its being true make in any one's actual life? How will the truth be realized? What experiences will be different from those which would obtain if the belief were false? What, in short, is the truth's cash-value in experiential terms? (Soames, 1984, p. 212)

He continues to elaborate how truth is judged based on how relevant it is to one's situation. If a thought occurs an individual at a given point in time, then it will be adopted or discarded in relation to how significant that thought is in that moment. Thus, this theory holds truth itself to simply be a sort of "function" of usefulness, which is more consequential. The last theory of truth I will address is the one formally championed by Alfred Tarski, and the one this thesis adopts. Tarski (1944) argues that truth is essentially a biconditional, one-to-one relation, between a proposition and state in the world. If an empirically-based "the rose" satisfies x in "x is red", then "The rose is red" is true; for complex sentences, Tarski is the one who proposed the use of logical operators and quantifiers to verify the satisfaction conditions of the proposition, as natural language is prone to vagueness and ambiguity. A crucial point to make is that Tarski emphasizes that a metalanguage only applies to one object language, and thus a definition of truth only applies to the language being analyzed due to how that language's atomic expressions and predicates function and combine. My framework attempts to provide more universally applicable representations of concepts that apply to linguistic expressions uniformly via an ontological language.

Before ending this section on formal semantics, it would be wise to include one of the main foils to its approach; that is, semantics that denounces the use of an artificial metalanguage. In a convincing paper, Goddard (2008) claims that the study of linguistic meaning is best achieved via natural language itself, an approach he dubbed Natural Semantic Metalanguage (NSM). In this semantic framework, Goddard and Wierzbicka propose the use of semantic primitives that apply universally to languages as the atomic level of meaning that cannot be decomposed further and out of which more complex meanings are built. These semantic primitives are lexical entries, with common examples being categories such as "THINK", "WANT", "HAPPEN", "DO", "BEFORE", "FEEL", "BECAUSE" ... that function as umbrella terms for the lexical entries that fall underneath and as operators in

relation to other primitives. This approach is ideal, claims Goddard, as an artificial language of formal semantics strays too far from the “naturalness” of natural language and essentially becomes its own language that needs its own interpretative tools and relies on natural language itself to be properly understood, hence rendering its purpose moot. This is a compelling argument, yet is also compellingly foiled by the likes of Tarski and Frege, and all the following works that have been engendered thanks to their push for formalism as the stable approach to semantic analysis.

While this is far from a complete overview of formal semantics, it provides an accessible base upon which the arguments and formalisms of this thesis may be built, and the philosophical assumptions of which covered.

2.5. Modality and Deixis

In direct relation to this thesis, the linguistic phenomena of modality and deixis must be addressed. Regarding textbook definitions, Usmanova (2021) defines modal words as terms that express a particular attitude towards a proposition, which can be explained as the belief expressed in regards to the possibility, probability, or necessity of a given situation. On the other hand, Levinson (2006) defines deixis as contextually determined linguistic expressions that hold their meanings in subjective factors that are highly dependent on speaker intentions and locations of utterance. He also acknowledges how difficult of a task it is to attempt to treat these terms in an objective (formal) manner. It is clear how the general consensus towards modal and deictic terms in language are overwhelmingly pragmatic and dependent on human consciousness. A fair example of this would be Boyd and Thorne (1969), where they apply Austin’s speech act theory to various modal verbs in an attempt to provide a semantic analysis of these terms; this, I claim, is an unfounded approach, as Austin’s theory is a pragmatic one and not semantic, and much less formal semantic. If we explain modal verbs through the use of speech acts under the name of semantics, we are

essentially denouncing that they may be explained via isolated means or indicate anything about the world beyond speaker intentions; this is why their approach will not be tackled in my analyses, as I assume their theory to belong under a pragmatic consideration of modality and unrelated to my assessment of semantic theories. Hence, the issue at hand is the proper treatment of these categories in a formal semantic framework. There has been valuable work done in this regard and in addressing what modals and deictic terms truly are in a formal context, some of which I shall highlight now, as they are presupposed rather than refuted in my analyses. Cresswell (2006) points out that modal verbs, specifically boulomaic ones such as “believes”, “thinks”, and “wants”, are in fact attitude prescriptions for the propositions/clauses they precede and influence the truth-values of those propositions in unexpected ways. He illustrates how, for instance, a sentence like “It is raining” has a static truth-value: true if the present state-of-affairs (or the statement’s extension) matches the intensional meaning of the sentence, and false otherwise; however, if we attach a propositional attitude such as “believes” to the proposition and end up with “John believes it is raining.”, we notice how the truth-value of the full sentence suddenly becomes blurry, for it is not in virtue of the extensional-intensional correspondence or empirical validity that the truth-value is attaining, but rather in virtue of John’s relation to the previously isolated proposition (“It is raining”). The question is, thus, what exactly is this relation, and can it be accounted for in non-psychologistic terms? Moving further, to address the issues of necessity and possibility (found in alethic and epistemic modals), Kaufmann, Condoravdi and Harizanov (2006) point out that is, to no surprise, necessary to use possible worlds semantics, such as that introduced by Kripke, in order to account for them formally; this has been done before, as I will address in my analyses, yet attempts still prove distant from a fully objectivist framework, as we shall see in Kratzer (2012). Regarding deixis, Levinson (2006) notes that in Montague’s original works, deixes were simply “reduced to a set of indices for

speakers, addressees, indicated objects, times and places” (p. 15) and integrated into the intensions accordingly; this is a view I accept, as indexes are characteristic of the world itself rather than mental entities. Alternatively, Perry and Barwise (1983) expound on Situation Semantics, which is another formally sound approach that, while recognizing the existence of mental representations, makes the crucial distinction between what is cognitive and what is objective. In their words:

Human cognitive activity categorizes these situations in terms of objects having attributes and standing in relations to one another at locations-connected regions of space-time. Human languages reflect (and enhance) this cognitive activity by giving us a way of communicating information about situations, both those we find ourselves in and those removed from us in space and time ... we recognize the epistemological primacy of situations, but follow the lead of language and take objects, relations, and locations as the primitives of our theory, reconstructing situations from them. (pp. 668-669)

This approach of positioning objects, relations, and locations as semantic primitives for handling attitudes and spatial-temporal constraints is one I closely follow in my own terms. Despite this, they do state that their theory places mental realities as primitive points, and this is where their theory and mine diverge, as in my philosophical grounding, epistemological representations are formed only after ontological ones are recognized as fundamental. It is Kaplan’s (1989) widely adopted view of speaker-determined characteristic functions that I denounce in my analysis of deixis, as my definition of “context” refers to world-states that merit representation, rather than properties conferred to a situation by a speaker.

While these semantic investigations are to be considered foundational to understanding modals and deictic terms, we will notice in the upcoming analyses how many

consequent attempts at a formal semantic analysis of these categories fall short of describing ontological world states and rather fall prey to subjective variables, such as contexts and mental states, which depreciates their logical stability. This is what leads us to our polemic that modality and deicticity can, indeed, be defined and represented in a purely theoretical, semantic framework that stays faithful to the formal and ontological commitments of Tarski, Frege, Montague, and formal semantics as a field.

Finally, the exact categories that are tackled in this thesis are best laid out in their theoretical foundations here. The researchers that are critiqued in Chapter 4, in the respective categories they are attributed to, are presented below. This serves both as an overview of the modal and deictic categories I address, and an overview of what the researchers themselves that I assess have to say about them.

2.5.1. Alethic Modality

The most philosophically rooted of all modal categories, alethic modal verbs center around the linguistic units “always” and “sometimes”, along with all their lexical or phrasal equivalents, such as “It is necessary that...”, “It is possible that...”. What I aim to address is a strong background on what it means for something to be necessary or possible as expressed in language in the ontological sense, as one must also note that it is impossible to discuss one of those concepts and not the other. There is a particular lack of semantic representations for metaphysical necessity/possibility in the literature, and so I will aim to provide one that applies in continuity with the rest of my categories. For my philosophical grounding, I shall overview the works of Aristotle, Leibniz, and Quine on their original conceptions of necessity and possibility, followed up by a discussion of Marcus as it relates to my proposal.

One cannot discuss the notion of necessity without mention of Aristotle’s metaphysics on necessity, contingency, and their relation to Being and essence. In the *Metaphysics*,

Aristotle deliberates that necessity is identical with the highest order of Being, and only Beings that are independent of all other metaphysical or physical attachments (contingencies) may be called necessary, as they only exert influence and are never exerted upon; in other words, they are the “first” causes. All other beings that fall under the highest essences may be called contingent. Thus, his sense of necessity is a highly retrospective one, as it attempts to identify the purest of essential beings in its identification of necessary entities, and those qualities that have emerged only as accidental qualities (The Metaphysics 1015b19-26). This is why Aristotle does not subscribe to a causal theory of determinism, as “the accidental arises not by a process but instantaneously, as the effect of concomitance of heterogeneous causal chains.” (Breuer, 2017, p, 281), which means that events simply “occur” without a predetermined driver. Here, my theory diverges from Aristotle’s, as I hold that events do indeed emerge as part of a process that shall be named and represented below and are not simply accidental occurrences. On the other hand, Leibniz understands necessity as a precursor for non-contradiction determined and proven in a countable number of expositions, and possibility as when this fails to apply (Pickup, 2014). This is a basic notion that is accepted in this study, as I do aim at defining necessity and possibility in a computable manner that can be applied to finite natural language expressions; as per Leibniz, if a notion takes an endless number of formulae to be proven as contingent, then it is contingent by the very virtue of this fact. I also find no reason to disagree with this, as my representations of contingency are only derivations of my formulae for necessity, and it seems impossible to account for a contingency in a precise, formulaic form while accounting for every possible variable in a given world. Crucially, as will be relevant to this thesis, Leibniz notes that necessity formulations are finite as they always end in analytic identity statements, while contingencies do not, and we shall find that my analysis will yield a similar result. This is why my theory can be said to build upon that of Leibniz’s notions. Lastly, in opposition to

Leibniz, Quine (1948) argues that identity relations fall apart under modal contexts; if we take “Robert = Mammal with arms and legs” and “Robert = Baseballer” as a given identity statements, then under substitution law, “Mammal with arms and legs = Baseballer” should necessarily be true. Of course, this is not the case, and Leibniz’s substitutivity principle appears to fail, leading to Quine’s primary argument that modality cannot appear under the rules of standard logic, lest the system fails. This is why Quine seems to think metaphysical essentialism cannot be escaped if one is to do modal logic, as logical rigor alone cannot account for their peculiarity and a metaphysical grounding will be needed, notwithstanding its instability.

Marcus (1967) opposed Quine by distinguishing between two forms of essences. Moreover, he argues that we ought to be careful in characterizing necessary properties or states as “essential” in the Universe or to the specific entity as per Aristotle’s definition (Ballarin, 2024). For Marcus, necessity and possibility emerge in either a weak sense or a strong sense. Weak essentialism dictates that an attribute yAy exists necessarily only in select objects and not in others, while strong essentialism mandates that it is a necessary property in select objects and a contingent property in the rest. The reasoning behind this is that necessity is determined via the domain it occurs in, with these domains, unlike necessity itself, possessing a shifting and dynamic nature that varies across worlds. She lays out this argument with the following representations, where yAy stands for a necessary property for weak and strong essentialism respectively:

(2.5.1.1.) (WE) $(\exists x) (\exists z) (\Box (x \in yAy) \wedge \neg \Box (z \in yAy))$

(2.5.1.2.) (SE) $(\exists x) (\exists z) (\Box (x \in yAy) \wedge (z \in yAy) \wedge \neg \Box (z \in yAy))$ (Ballarin, 2024)

We may see how in (WE), entity x contains the necessary property yAy necessarily and entity z is not constrained by this necessity. In (SE), both entities x and z necessarily

contain yAy (strictly identical), yet z is precluded from complete necessity. She eventually states that weak essentialism is a minimal requirement in discussing necessities and identities, while strong essentialism becomes dubious when we realize that we may substitute every occurrence of an entity in a formula for its indiscernible equivalent and end up with the same interpretation *ad nauseam*. However, if we are to disregard metaphysical vagueness, strong essentialism is still a plausible interpretation. The stability of this dualist approach to essentialism, and by extension, whether or not such a division is even necessary to discuss necessity is assessed in Chapter 4.

2.5.2. Deontic Modality

I now turn to the second type of modality, namely deontic modals, which is less philosophically grounded compared to alethic modals and fall under a linguistic umbrella for the most part. I shall similarly overview a handful of historical approaches to deonticity and selectively adopt or critique them as I build my own framework. As a solid beginning, Von Wright (1951) provides a clear standard definition of what constitutes deontic modality, claiming that they are those categories that exemplify the notion of obligation and permission, and that “It is of acts and not of act-individuals that deontic words are predicated.” (p. 2), meaning that deontic modals always and only quantify over general acts and not specific acts done by specific individuals, which I assume falls under the former idea as a given. Both of these notions I consider fundamental and indisputable in my system; it is what comes next, however, that I will dispute. In his paper “Deontic Logic”, von Wright (1951) attempts to establish deonticity on the same truth-conditional, propositional grounds as alethic modality by mapping the rules of formal logic unto his new deontic operators, such as P (permissible) and O (obligatory) and essentially treating these notions as variations of formal logical rules. While the mapping is relatively successful, he acknowledges that the very nature of obligation is problematic and resists complete formalization in the same degree

as alethic modals, commenting that “It might, however, be questioned whether it can be regarded as a truth of logic that a tautologous act is obligatory.” (p. 10). Indeed, a strong opposition to the alethic formalization of deonticity can be found in Ross’ Paradox. As expounded by Hansen (2006), Ross’ Paradox throws a wrench in the logical understanding of obligation by stating “If logic states that “A is always true implies (A $\vee$ B) is always true”, then if mapped onto obligation, “If O(mail the letter) is always true, then O(mail the letter or burn the letter) is always true.”. This is clearly a complete mismatch between what we expect holds of obligation in natural language and what logic dictates should hold, and this occurs due to the commitment to the selfsame formal logics that are applied to necessity and possibility. Nevertheless, this commitment is further established by Kanger who, according to Hilpinen (2001), mapped deonticity to Kripke’s possible world semantics and tied it to concepts such as agency, counterfactuals, necessity, sufficiency, and certain categories of “right (to do)” in terms of formal definitions; I shall not be outlining all of Kanger’s formal systems here due to the limit of my scope and because my issue concerns Kanger’s commitments rather than his formulations themselves. Just one example of his deontic analyses that I find highly relevant concerns the following found in Hilpinen (2001):

(2.5.2.1.) $u \models D_S(a,p)$ if and only if $w \models p$ for every w such that $S_{DS}(w, u)$

(2.5.2.2.) $u \models D_N(a,p)$ if and only if $w \models \sim p$ for every w such that $S_{DN}(w, u)$ (p. 141)

Taken separately, the operator-function $D_S(a, p)$ means that an action undertaken by a is a sufficient condition for p to occur, hence it is not necessary for a to act, and p may still exist otherwise in a counterfactual situation (making a ’s act for p not obligatory). For $D_N(a,p)$, an action undertaken by a is a necessary condition for p to occur, and in case a does not act, p cannot occur counterfactually. Lastly, $S_{DS}(w, u)$ and $S_{DN}(w, u)$ signify that the worlds w and u reflect each other in a ’s actions via an accessibility relation under the sufficiency and

necessity conditions respectively. With this reading, we may understand that (2.5.2.1) means “p occurs in every world w accessible from u iff the action that a performs in u is sufficient to bring about p in the model” and (2.5.2.1) as “p does not occur at every world w accessible from u iff a does not act in u.”. These models make us see that obligation is tied to counterfactuals and possible worlds in order to be comprehended (at least in Kanger’s system), as it is the reality of an event occurring via a required action that drives the obligation system, a process Kanger calls “agency”. Indeed, this analysis is quite sound, but what is missing from Kanger’s formulation is what exactly necessitates an action itself to occur, as this system only shows us what obligation looks like across a span of worlds and how it is tied to events, but not what exactly connects possible actions to the occurrences of their respective events.

Kratzer’s approach to obligation is another theory that merits mentioning, as I find some issues within it. Let us first, however, outline her framework. Kratzer (2012) states her well-known account of how deontic modes of obligation function in language, and firstly provides the following summary for her initial concept: “A deontic conversational background is a function f such that for any world w , $f(w)$ represents the content of a body of laws or regulations in w .” (p. 37). Beebe (2017) paraphrases the notion of Kratzer’s (2012) conversational background by explaining that it contains all the socially and individually held normative judgements against which the elements of a conversation are evaluated. Secondly, this conversational background established what she called a modal base, which is a set $F(w)$ that examines which accessible worlds from the current world, in which the conversational background is established, are also compatible with or allow that background to be true (Beebe, 2017). Lastly, and the most crucial component of Kratzer’s framework, is her use of an ordering source mechanism, which she illustrates as:

(2.5.2.3) For all worlds w and $z \in W$: $w \leq_A z$ iff $\{p: p \in A \text{ and } z \in p\} \subseteq \{p: p \in A \text{ and } w \in p\}$ (Kratzer, 2012, p. 39)

For an intuitive definition, she defines this formula of the ordering source as the set of all worlds that support the conditions of the modal base in which the conversational background is situated as closely as possible, and ranks these worlds by degree of similarity to the given base. Formally, then, the formula would state: The worlds w and z belong to the set of all worlds, the world z contains a modal base (A) formed of a set of propositions that is evaluated against world w , and the ordering source “attaches” world w to z if and only if both worlds share the domain that the modal base of w possesses or are otherwise identical in those domains. For Kratzer (2012), this ordering mechanism, when applied to deonticity, could serve to account for the notion of obligation. If we take the modal base $F(w)$ to contain all the normatively determined propositions or acts that are accessible in a world, then a statement is obligatory if all the closest worlds captured by the ordering source mechanism also necessitate its obligatory nature in virtue of their shared background of normative premises. Otherwise, if no worlds or relatively few worlds are ordered as such, then the statement lacks obligatory modality. For instance, the ordering source will search, identify and rank the possible worlds where statements analogous to “Parental figures are to be respected” hold true to determine the obligation value of “Peter must obey his father” in a given world. If all best-ranked worlds contain the first premise as a normative truth, then the latter qualifies as an obligation. This is the gist of Kratzer’s theory on how modals function in language – a topic of central scrutiny in my analysis of deontics, as it appears to suffer from both semantic and pragmatic oversights. A last study worth citing is Belnap (1990), as he came to the realization of highly similar conclusions and a proposition that I accept. He observed “The grammatical fact, and one that should guide us philosophically, is that obligation and permission and prohibition always take imperatives.” (p. 22). For this, he

offers us the argument that the standard notations and understandings of O and P (obligatory and permissible) do not account for their embedded clauses linguistically, for those clauses are in fact imperatives and not declaratives, which he represents with Obligated[α stit: Q], which stands for α is obligated to see to it that Q. This is a better alternative to the O operator as it does not commit what he calls the “Declarative Fallacy”, which is a common fallacy in logic and philosophy that gives special privilege to the analysis of declarative sentences as somehow more representative of language. So, instead of saying obligation operates over a declarative phrase, we ought to say it, instead, operates over an imperative one. This is a view that I appreciate philosophically, as my own view assumes that obligation functions locally and with very much the same force as an imperative, albeit with different operators, as detailed in Chapter 4.

2.5.3. Boulomaic Modality

As the last modal form I intend to tackle, it is worth noting that boulomaic modals are quite divergent in comparison to alethic and deontic modals, as unlike those two, it is quintessentially rooted in mental states in almost every standard interpretation. Kiefer (1987) defines boulomaic modals as expressions of wish and desire upon propositions, which is why many have come to call these expressions propositional attitudes. An influential treatment of attitudes can be found in Hintikka (1962), wherein he provides an initial introduction of possible worlds to the notion of attitudes. In brief, his central claim was to map world accessibility relations unto knowledge and belief states belonging to a subject. He argued that “x believes y” is true in a world w iff x’s beliefs are compatible with a given set of worlds w’. This might appear to be a precursor to Kratzer’s (2012) ordering source mechanism, as once again, worlds are selected according to how tolerant those worlds are of x’s beliefs in w. In fact, this is one of the first treatments of attitudes in a manner that incorporates a possible worlds framework.

We move next to Portner and Rubinstein (2020), who provide a more tailored analysis of my subject-matter. To understand their overview of desires, one must define three essential terms historically provided by them: Simple necessity, human necessity, and diversity presupposition.

- a. Simple necessity is a basic indicator that the proposition p is true in all worlds where the subject's belief of p is true. (Originally given by Hintikka)
- b. Human necessity is essentially the ordering source framework for desires. It ranks the best worlds according to whether the subject's desires point to worlds in which the desire within the proposition is necessarily true. This constitutes the truth-condition for the propositional attitude of desire.
- c. The diversity presupposition is an assumption that one cannot want something either guaranteed or impossible.

Given these assumptions, Portner and Rubinstein provide the following approximation of the truth value of $[[\text{want}]]$:

$$(2.5.3.1) \quad [[\text{want}]] = [\lambda p \lambda a \lambda s: a \text{ is the experiencer of } s \wedge \text{hn}(p, \{\text{bul}, \text{dox}^+\}, s)]$$

This reads as: For a proposition p , agent a , and situation s : For the best-ranked possible worlds, ordered by *bul* according to the modal base dox^+ both determined by hn (human necessity), the proposition p held by the agent is true in s . In other words, this is a precise formulation that tells us how desire is tracked across worlds according to how an agent's desires view those worlds as “desirable” based on his/her beliefs. This is a steadfast analysis, as it sets clear parameters on how these worlds are identified and accounts for situations and backgrounds while doing so. However, it lacks a clear definition of what desire itself is and why it necessitates itself within a subject, or how it is teleologically related to an act that it may or may not engender. Portner also attempts a formulation to represent desirability in

primitive terms: “*short-term-pleasure-bouletic_{addressee}*” (Portner, 2007, p. 374), which may simply provide us a preliminary understanding of what boulomaics are in terms of natural language primitives, yet it is crucial to remember that these primitives are not formal and cannot be claimed to be logical, as pleasure is a psychological reality, not an ontological one. Thus, this line of analysis does not decompose the concept of “boulomaic” sufficiently enough and is relatively satisfied with treating it as a form of predicate across worlds.

A last point of view worth considering before moving to my framework belongs under the domain of dynamic semantics, but is one that explicitly incorporates emotive states of the subject. The framework of MSDRT (Mental State Discourse Representation Theory), championed and derived by Hans Kamp’s Discourse Representation Theory (DRT), integrates a mentalist framework for representing meaning, one that is highly reminiscent of Locke’s view on ideas being the root of meaning (Kamp, 2022). While the inner workings of the syntax and structural rules of a DRT system is beyond my scope, its primary assumptions may be overviewed and one of its structures tackled. Kamp explains that the primary purpose for the inception of DRT was to account for complex cases of anaphora, such as the famous donkey anaphora problem, by framing meaning at the evolutionary discourse level rather than statically as most semantics does. Even possible worlds semantics, he claims, is insufficient in capturing how a mental state such as belief is attributed to a subject and simply transforms these reactive, human notions into inhuman, rigid sets across worlds. DRT, and by extension, MSDRT accounts for meaning by tracing them along a discourse, often across sentences. This is done by designating ERs (entity representations) to individual referents, tackling emotive states as attributive predicates, and holding that a discourse takes place in a particular domain in which the ERs possess transitive identities across different levels of the discourse, which explains discourse coherence. and as seen in the basic example given by Kamp (2022, p. 301):

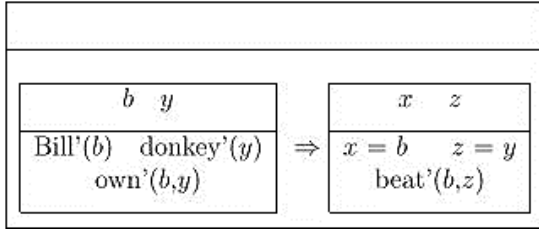


Figure 1: Example of Kamp's DRT system

It is apparent how Kamp accounts for conditionals and anaphora by representing entities (via ERs) in their own domains and transitively equating those entities as the discourse progresses and noting the conditional operator between the phases. This provides a clear visual of how discourse referents evolve in a discourse setting. Kamp (2022) notes that DRT alone has a debatable status in its commitment to psychological states in semantics, with proponents advocating for both interpretations. However, MSDRT evolves this framework to explicitly account for psychological states within the agents of discourse. This can be seen in the following example, derived from Kamp's work, that structures the sequence "The cup I am holding contains coffee. I hope that it's still warm.":

$$\begin{aligned}
 (2.5.3.2) \quad & \{ \langle \text{ANCH } x, [x:\text{cup}(x), \text{hold}(i,x)] \rangle, \\
 & \langle \text{BEL}, [y:\text{contains}(x,y), \text{coffee}(y)] \rangle, \\
 & \langle \text{DES}, [:\text{warm}(y)] \rangle \} \text{ (Geurts, David, \& Emar, n.d.)}
 \end{aligned}$$

This structure first provides an anchor in which the external facts are laid out, then layers unto it two emotive states (belief and desire) that develop the propositional content found in the anchor. Furthermore, as a more complex example, Kamp (2022, p. 307) provides us the following structure in Figure 2 to account for a subject's belief towards their own propositional attitude statement with the example: "Fred believes that Julia is in Paris. He would like her to be in Berlin." This box structure may be interpreted as, first of all, a set of ERs belong to a speaker S for the present entities in the discourse (namely Fred, Julia, Paris,

and Berlin), all of which are assigned an external-to-discourse referent such as $f_S, j_S \dots$ in their own discourse representation structures K_x . As the argument to this first set, we are given the attitude set containing entities and indicators of emotive modes (BEL, DES). After these are all given, the meaning of the sentence becomes clear, as the entity denoted by K_{Julia} is juxtaposed with the entities denoted by K_{Paris} and K_{Berlin} in the modal functions of BEL and DES. Crucially, the referents denoted within the second box (within the MSD: mental state description) always map unto the ERs represented by S in the upper half of the structure, which denote real entities. Of course, this is a highly succinct exposition of Kamp's work and does not do the complexity of his framework justice, but it does serve to provide us with hints towards his foundational assumptions and semantic commitments that I shall forthwith examine.

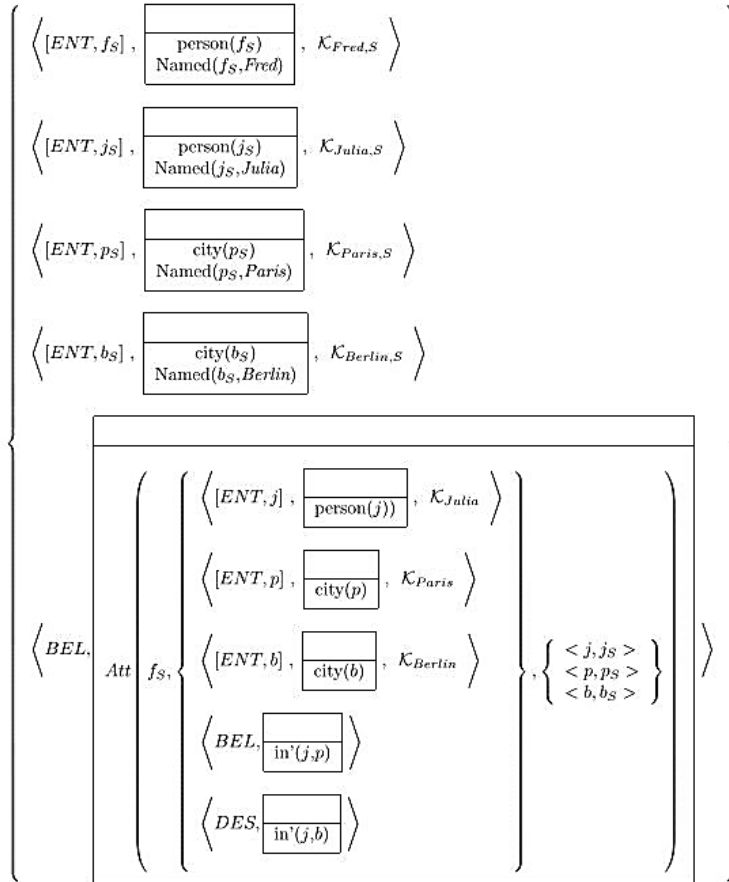


Figure 2: Example of Kamp's MSDRT system

2.5.4. Deixis

Following the original exposition in 2.5, Levinson (2006) defines deixis as “inadequate descriptions” (p. 11), which is, for the most part, an adequate description. Deictics pose a completely different, yet not less troublesome, challenge from modals, as they are not indicators of mental states but are rather indicators of context that are symbiotic to their linguistic environment. This makes it particularly difficult to ground deictic categories in an isolated, formal grounding as I did with modals, for, as Levinson stated, they only describe partially. As a brief note before addressing spatial deixis, let us take a look at Frege’s (1956) view regarding this issue, as he himself struggled with the notion of indexicals in his formal endeavors, explicitly remarking:

The case is the same with words like " here " and " there ". In all such cases the mere wording, as it is given in writing, -is not the complete expression of the thought, but the knowledge of certain accompanying conditions of utterance, which are used as means of expressing the thought, are needed for its correct apprehension. ... The same utterance containing the word " I " will express different thoughts in the mouths of different men, of which some may be true, others false. (p. 296)

Indeed, the way in which I address the phenomenon of deixis follows Frege’s conviction closely in that they stand for “conditions of utterance” as they are backgrounded within speech. What Frege states regarding the personal deixis “I” holds true just as well with spatial and temporal cases, as we shall soon see. It will thus be my own endeavor to lay out what these conditions of utterance in the world are as they relate to the category of deixis they represent.

2.5.5. Spatial Deixis

It seems feasible to first investigate the philosophical inquiries of G.M. Moore (1918) regarding the notion of identification in reference, as it will also be a problem in my own formulations. While discussing the issue of how we contextually refer to different entities, Moore brings up the metaphysical conundrum of “parts”. He argues that all objects are necessarily comprised of what is only appropriate to be called its parts, as used in common speech, even though what exactly constitutes a part of an object is never a thing of clarity. It is this lack of clarity, he claims, that makes references to objects particularly troublesome, as we have no clue as to what limits the range of our reference when we refer using language. To elaborate Moore’s point, we have little knowledge of what exactly it is we refer to in space when we say “this x”, as “this” always seems to refer to a metaphysical constituent of x and never the whole nature of x itself; for instance, the utterance “this apple”, will likely refer to what empirically forms the surface or insides of an apple, yet an apple contains far more properties or predicates than its surface and core, and so, they are filtered out, it seems, by our act of reference in favor of a more simplistic, conceptually-determined idea of an apple. Hence, what determines the reference of x, depending on one’s philosophical stance, will either only refer to one specific component of x or refer to everything that takes x into itself. Both of these views are clearly problematic and fail to reflect what we truly do when we refer, although most of the literature in philosophy tends to favor the former interpretation, as the latter loses all sense of containment. This is a mental problem as a whole, I believe, but it does cross over to deictic expressions and makes it particularly difficult to abstract into world-states when a cognitive constraint of such a degree constantly lurks in the analysis. I will leave Moore’s concerns in the background for now, but we shall see its relevance arise in a newer form while constructing my framework. I now crucially move to the main figure

whom I assess in deixis research and, as I did with modals, assess its formal stability going forward.

Kaplan (1979) presents an argument that in continuing Frege's commitment to the idea of senses, we may furthermore divide them into two types: Content and character. The content of a sentence is the proposition expressed by it, or rather, the extensional side of its meaning that is also crucially dependent on the context of the proposition, yet it is the idea expressed by it nonetheless. The character of a sentence, however, is a function from contexts to content; it decides what interpretation the content will be given depending on the context in which it is spoken. This is why mention of character is seldom seen beyond the discussion of indexical notions, as the character of a proposition will be irrelevant if its meaning is insensitive to context changes, for it will be true in all contexts. To expand further on this, let us also consider Kaplan's basic parameters by which he evaluates truth-values: $\mathcal{U} = \langle C, W, U, P, T, I \rangle$ (p. 88), where C stands for the set of all contexts defined by "If $c \in C$, then $\langle C_A C_P \rangle \in I_{Located} \langle C_T C_W \rangle$ " (p. 89), W stands for the set of all worlds, U stands for the set of all individuals, P for the set of all positions, T for the set for all times, and I for the function that maps times and worlds to individuals and positions. Lastly, f will be taken to be an assignment of either individuals or positions to variables, defined as: $\exists f_1 f_2 (f_1 \in U^{V_i} \& f_2 \in P^{V_p} \& f = f_1 \vee f_2)$, which states that f is either an individual or position. Thus, in Kaplan's standard analyses, he makes use of the form:

$$(2.5.5.1) \quad \models_{cftw}^U \phi \text{ iff } \varphi_{cf}^U(tw) = \text{Truth (p. 91)}$$

This implies that any formula in a particular context is entailed to be true if, under the domain of $\mathcal{U}$, the proposition ϕ as it appears in context with a certain assignment is true at a time t in the given world w . When it comes to character, it is precisely that which assigns a proposition ϕ its $cftw$ parameters in a context based on the structure of $\mathcal{U}$. However, Kaplan (1979) does

not provide an explicit formula on how precisely this function of character assigns cftw to a proposition, and prefers to provide stability relations of character to identify when the context of an utterance changes or remains static, along with their own structures that are not directly relevant for my issue. There appears to be a need to provide what it is exactly about character that interacts with the set $\mathcal{U}$, or in other words, “what is the structure of character?”.

Moreover, upon contemplation, the set of $\mathcal{U}$ can be reduced to the amalgam interaction of contextual environments and possible worlds – a point I elaborate in Chapter 4.

My second point of reference will be the work of Gareth Evans regarding specifically the notion of demonstratives, although, like Moore, his work is more grounded in philosophy than formal semantics. In the general terms of his view, Evans (1991) regards the main contributor to the meaning of deictic propositions to be an “information link” between a subject and object, which can be defined as a continuum of information that an agent apprehends of a specific object as it moves through space and time, thereby forming an Idea of that object in the sense that it may be referred to deictically. This Idea is certainly mind-based, and constitutes a descriptive state of knowledge regarding the given object. In the case of a more superficial attempt at reference for objects beyond immediate reach, Evans (1991) argues:

The truth of an identity-proposition of the form $\phi = \text{that man}$ (where 'that man' is used of a man seen on television or heard on the radio) can consist in nothing but the fact that ϕ is a fundamental Idea of a man who is causally responsible for the sounds or images which the subject is perceiving; and so a knowledge of which man is in question can exist only in the presence of the idea of tracing the immediate objects of perception back to their causal source. (p. 149)

It appears obvious that Evans' framework clearly accounts for time, space, and causality for how we refer to objects via indexes. For now, I partially support Evans' views in that I agree that time, space, and causality are most certainly factors that determine deictic reference; however, I intend to add to this notion that these structures are modelled in the world before they are apprehended by a subject and turned into descriptive information. Regarding spatial deixis in particular, he argues that they are determined by particular "frames of reference" (p. 151) that comprise spatial connections among objects in that frame. This is a claim most supported by this thesis. Evans continues to state that these frames are related to what he calls cognitive maps of spaces that help determine reference. It is unclear whether he means to say that these frames are the cognitive maps themselves, or whether the frames give rise to the cognitive maps. As for this research, I shall take the latter claim to be in accordance with my standard view. Finally, most of Evans' discussion concerns the notion of egocentricity in his account for the index "here", in which he accepts the notion that "here" essentially refers to the place in which the subject is found, but grounds this place in the subject's perception: "... finally, and crucially, the subject is able, upon the basis of the link, to locate the object in egocentric space, and thereby in objective space." (p. 174) This seems to be a precarious conclusion, as it is vague whether the objective space merits an independent existence beyond the subject's cognition of it. Nevertheless, I do respect Evans' ideas on how the spatial "here" is accounted for, for even if he does not provide formal systems or clear verdicts on what a frame of reference or objective space is, he still identifies them as crucial elements in what the unit "here" signifies. All three approaches that have thus been covered suffer from the very same issue of vagueness in formulation, as it is understandably difficult to situate a notion commonly thought of as a concept in the world itself prior to cognition.

2.5.6. Temporal Deixis

As modes representing time, it should appear clear to us how this category of deixis can be smoothly accounted for in my LS mechanism, which already makes fair use of temporal parameters. However, while we are not as constrained by space as in spatial deixis, we might be just as much be troubled by the “contextual” nature of time indexes such as “Now” and “Then”, which will be my tackled notions. I strive to account for these linguistic umbrellas in a manner that belongs to the world’s structure LS. I do not include Kaplan this time around, as his theory of deixis in 2.5.5. provides the selfsame treatment for spatial and temporal deixis alike, and so his inclusion would be redundant. Rather than any stark criticism, this section differs from the rest by singularly focusing on the framework of Reichenbach, who suitably provides a formal analysis of temporal deixis that I critique and add to from my own formal resources in Chapter 4.

One of Reichenbach’s (2005) most notable ideas is the notion of reference time. This mechanism is what allows a deictic reference of time to “scale” or move towards another point of time than the time of utterance. He provides two other types of time reference as well: Speech time and event time. Speech time is the moment of deictic utterance when reference begins and the reference time is consequently designated, while event time is taken to comprise the various, specific events that take place within the reference time. He uses these three categories of reference as variables that shift across time to account for the meanings of different tenses. This is a philosophically sound perspective, as the three parameters he makes use of avail from clear distinctions and locations in space-time – a view that my own framework appreciates. Nonetheless, it may stand to benefit from proper formalizations into a rigid ontology and an address of causality in its categories, an undertaking I attempt by formalizing Reichenbach’s system and integrating it into my own.

As such, the theoretical background and the researchers directly relevant to this thesis have been investigated. In the analytic chapter, what has been discussed here is taken as a collective given and critiqued in the light of new frameworks and theoretical bases that I construct, ones in greater favor of semantic formality and ontological stability.

Chapter 3

Methodology of Research

3.1. Research Design

This thesis adheres to a qualitative approach, as it is dependent on philosophical analyses as its grounds for argument. Moreover, a descriptive-analytic pattern is followed in this thesis with some elements of a historical design. In reference to the review of literature, previously laid semantic analyses in each category are set forth and delineated in sum, which will be reflected upon according to a new semantic commitment and countered with alternatives that better uphold formal commitments.

3.2. Corpus

The theories assessed in this thesis may be referred back to the writings of various linguists, philosophers, and formal semanticists. Primarily, this thesis will take Frege, Tarski, Russell, Carnap, Kripke, and Montague as the theoretical foundations of both my goals and analyses along with the adaptation of their formal methods for my semantic analyses. The theories under analysis primarily include the works of Barcan Marcus, Georg Henrik von Wright, Stig Kanger, Angelika Kratzer, Nuel Belnap, Jaako Hintikka, Paul Portner, Hans Kamp, G.E. Moore, David Kaplan, Gareth Evans, and Hans Reichenbach. The works of these researchers were selected as they offer some of the most in-depth formal semantic analyses of modality and deixis to date. Scholars like Kanger, Kratzer, Kamp, and Hintikka offer different approaches on how to tackle modal expressions in language formally, with methods such as possible worlds analyses, ordering source frameworks, and Discourse Representation Theory (DRT), which attempts to account for psychological phenomena. Kaplan and Reichenbach provide fascinating perspectives on how to formally handle deixis as used in language with their different theories of reference found in space and time. All of the above

theories and methods are either partially adopted, adapted, or criticized according to their adherence to the more rigorous objective commitments of this thesis, and the degree of their relevance to it.

3.3. Procedure of the Study

This thesis is divided into two major sections, pertaining to modality and deixis respectively. Each of these sections are further divided into sub-sections belonging to the corresponding types of modals and deixis, with Alethic, Deontic, and Boulomaic modalities and Spatial and Temporal deixis addressed in that order.

Within each sub-section, two divisions are found, for descriptive and analytical purposes respectively. In the descriptive half, a relevant researcher who has proposed a well-known theory relating to that sub-category is cited according to their expositions in the literature review. In the analytic half, two different yet interrelated methods will be used: Philosophical and Logical-Semantic, which is the pattern of analysis I adopt systematically throughout the study.

After all the analyses, the resulting formal logical representations of all the addressed linguistic categories are laid forth in table form.

3.4. Data Analysis

After the corresponding theories are laid forth in each sub-section and the analysis begins, the pattern of analysis is as follows.

1. The philosophical implications and commitments of the given researcher's theory or framework will be assessed and critiqued according to my own overarching philosophy of formal semantics, which prioritizes objectivity and freedom from cognitive and contextual constraints.

2. Upon rigorous consideration of the above-mentioned philosophical inquiries, the resulting ontology of the given linguistic category will be used to create a formal logical representation of meaning. This will be done by transforming my philosophical categories pertaining to that linguistic domain into a model-theoretic formal language, and the consequent combination of those given symbols into formulae that represent and capture the meanings of the assessed linguistic category.

Chapter 4

Analysis

4.1. Alethic Modality

My first matter of contention concerns primarily the work of Barcan Marcus as delineated in section 2.5.1. of the review of literature. As was discussed, Marcus' (1967) work was a critical buildup from previous propositions regarding the nature of necessity, as she offers a specific metaphysics to account for the nature of necessity and possibility across worlds. Leibniz had proposed the idea that any formal-logical representation of necessity must be finite in its scope, while those of possibility extend indefinitely. Quine (1948), on the other hand, entirely denounced the use of formal systems in the endeavor of modal analysis in language. Marcus thus attempted to prove this wrong by suggesting a distinction between weak and strong essentialism (WE and SE) defined as 2.5.1.1. and 2.5.1.2. This distinction makes use of domains to maintain its explanatory powers, as a weakly essential property occurs in particular entities and is absent in others, unlike strongly essential properties that occur in all objects yet lack a necessary nature in some of them, thus being contingent. The central point in Marcus is that while necessity itself is proposed as universally occurring, the domains in which necessity presents itself are ever-varying. One of her purposes is to refute a purely prosaic, metaphysical explanation of necessity as provided by Aristotle, and oppose Quine's rejection of formal logic as applied on modal cases. As properties face the issue of substitutivity as discussed by Leibniz, Marcus' system faces the same issue, as it relies highly on the spread of properties across worlds and entities in both WE and SE. This makes her system metaphysically opaque in much the same way that Aristotle's definitions were. I shall attempt to present and avoid this issue in my consequent analysis of the problem.

My arguments support that of Marcus only partially. I do support her rejection of Aristotle's essentialist argument of necessity and agree that modality is best expressed formally as she has done. I also acknowledge that necessity and identity are congruous, but expect another form of proof that does not rely on indiscernibility as its presupposed base, nor on categorizing essences as a way of clarification, but rather on ontological states. This is because the notion of indiscernibility (as expounded by Leibniz) is dangerously analogous to identity, and to use indiscernibility as a foundation of proving identity between objects runs the risk of a circular argument, a commonly threaded ground by many philosophers. Indiscernibility is a valid notion only if it is axiomatically implied by or derived from a system, but not as an assumed basis for proving the adjacent notion of identity. Moreover, a weakness in her WE/SE categories is apparent. I propose that if:

- a. Weakly essential properties only occur in particular event-times (domains) and not in others, they are not to be deemed necessary in that world but rather as contingent.

Formally: This argument suffers from allowing a proposition to possibly necessary ($\Diamond \Box p$) at one world W_1 and possibly not necessary ($\Diamond \neg \Box p$) at another W_2 , thus it cannot be necessarily true in all possible worlds.

- b. Strongly essential properties are necessary in particular event-times and contingent in others, they are reduced to contingency. Formally: If property $F(x)$ necessarily appears in W_1 for x and contingently in W_2 for y , or $[\Box F(x) \wedge [\Diamond \Box F(y) \Diamond \neg \Box F(y)]]$, then this implies F is not necessary globally.

Hence, both formulae come to represent possibility and not true necessity in the sense of "true in every possible world", a definition that modal logic requires, and by extension, truth-conditions, require.

To resolve this, I have thereby formulated a certain system. I believe that necessity is not characterized by metaphysical essentialism at all, but by the logical structure (LS) of the Universe and a phenomenon I dub logico-structural inevitability (LSE). The world's events are drawn forward by a particular structural sequence that either allows or disallows their existence; it is not that objects contain essential or non-essential properties, but rather that properties are engendered in objects as this logical structure dynamically evolves. What I call essential properties are simply those properties that have structurally developed most frequently in this logical flux, but may have very well not developed in that particular way. Thus, the equation I propose to account for the semantic reality of necessity follows this argument:

Let:

- a. $\phi(\text{LS})_0$ represent the logical structure of the current world at its given state.
- b. $\phi(\text{LS})[\phi(x)]\psi$ represent the dynamicity of the logical structure of the world at any state, defined formally as the abstraction:

$$(4.1.1.) \quad \Box [W_\infty: \{W_0, W_1, \dots W_x\} \rightarrow (\exists y \in W_\infty) \wedge [y = \text{Event}(\text{Time})] \wedge \phi[y]\psi]$$

This translates to: *It is necessary that across the infinite set of worlds, a particular y exists in the span of worlds W_∞ , this y is an event as a function of time, and this y is a subject to change under the process of LS development.*

Out of this evolution, we draw:

$$(4.1.2.) \quad v(\text{Always}) = 1 \leftrightarrow [\phi(\text{LS})[\phi(y)]\psi] \rightarrow \psi[\forall x \exists y [W_x \wedge \text{Event}(\text{Time})_y]]$$

Where all y-s, characterized as particular updates upon the given LS, occur at every world W_x as determined by the development of the LS at specifically constant event-times in ψ .

$$(4.1.3.) \quad v(\text{Sometimes}) = 1 \leftrightarrow [\phi(\text{LS})[\phi(y)]\psi] \rightarrow \psi[\exists x \exists y [W_x \wedge \text{Event}(\text{Time})_y]]$$

Where all y -s, characterized as particular updates upon the given LS, occur at particular worlds W_x as determined by the development of the LS at specifically constant event-times in ψ .

The term “event-time” is used to designate a precise notion that encompasses:

- a. An occurrence of a particular state (event).
- b. A particular point in time.
- c. An ontological environ that is innate to the combination of a and b.

Point c is what is essentially the standard domain of a given event, but rather than varying across worlds and quantifying over events, they are necessarily innate to them. Domains occur when events occur at particular times and cannot stand in isolation, implying that what becomes updated in the LS is not c, but rather a configuration of a-b, with c remaining rigid across all worlds and what necessarily preserves the continuity of a-b into progressive states. As we saw with Marcus, properties alone run the risk of triviality by substitution due to how ungraspable they are by an ontology; conversely, an event-time is always grounded in a constant domain that ensures its survival in the system unto newer forms. One may formalize this as: $\Box \forall w [Event(Time)y \in D_y]$. This ontology of domains will be a perpetual commitment in the thesis.

Using the $\phi(LS)$ apparatus, we gain a clearer picture of how necessity and possibility come to be. The function $\phi(LS)[\phi(y)]\psi$ is interpreted as any given LS, $\phi(LS)$, undergoing a particular change (akin to Aristotle’s kinesis or Hegel’s antithesis) characterized by $\phi(y)$, which introduces a new, unprecedented element (event-time function) y into the former LS represented by ϕ . The function then outputs a new LS, namely ψ , which now contains the new property of y along with every property already contained in ϕ . With a certain mapping, a glimpse of the structural form for “always” to account for necessity would look like:

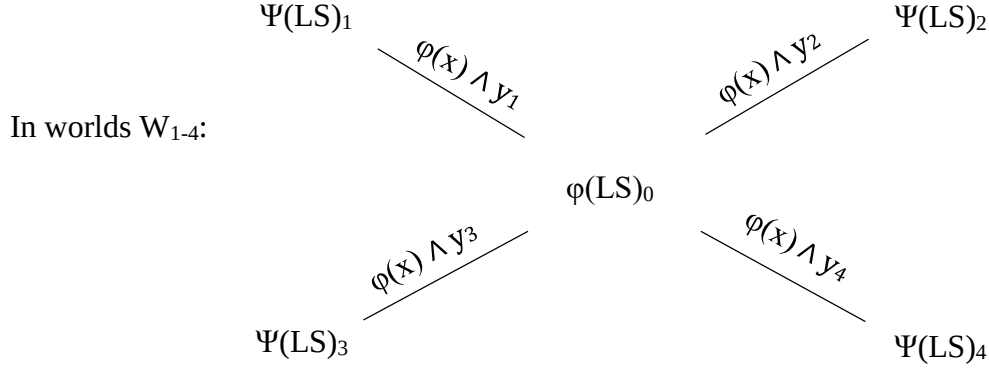


Figure 3: Structural representation of necessity in an LS system.

Wherein y represents Event(Time)s. This process repeats infinitely across infinite branches, as LSE determines that all specific event-time updates $\phi(x)$ will occur at every event-time y_x in every possible node, which will become an innate property of all new states $\psi(LS)_x$. However, it is crucial to note that this property is not “metaphysically innate” to the object, but only “necessitated by the LS flux” that led to its existence; this is what I name LSE. How this happens is that when a certain event-time y becomes actualized by the LS, it becomes ingrained in the system $\phi(LS)$, represented as:

$$(4.1.4.) [\phi(LS)[\phi(y_r)]\psi] \rightarrow \psi_r[\forall x \exists y [W_x \wedge \text{Event}(\text{Time})_y]] \rightarrow [\phi_r(LS)[\phi_r(y_u)]\psi] \rightarrow \\ \psi_{r,u}[\forall x \exists y [W_x \wedge \text{Event}(\text{Time})_y]] \dots \text{ where } r \text{ and } u \text{ symbolize unique event-times} \\ \text{brought in by the structural progress of the world and become embedded in its} \\ \text{consequent states.}$$

Thus, once an event-time occurs, it is much more likely that it will re-occur in the system; the more it occurs, the more likely that it will occur again. As this probability increases, so does the event-time’s probability of occurring simultaneously in every world W_x at one specific time interval (this is supported by the fact that moments of time are infinite, and so are possible worlds; thus a simultaneous occurrence in all W s at some t is necessarily a given). When this happens, the event-time becomes truly necessary, as it is now “promoted” and

ingrained in every possible world. For highly axiomatic statements, such as mathematical propositions, they are viewed as innate properties of the LS itself, existing in absolute identity to it. As to why the above phenomenon occurs (why y happened to develop frequently instead of another event-time), it is strictly arbitrary and based on probabilities, much like a number generator; some numbers generate multiple times and some are only seen once, and some hitherto not at all. Therefore, across infinite points of time and infinite possible worlds, if an event-time y universally occurs at all worlds, it is necessary in the LS for all of time. This essentially how I propose necessity is grounded ontologically, which is admittedly a strict definition, as necessity only applies to properties that have undergone the above promotion in the LS. The promotion principle for event-times in the LS goes as follows:

(4.1.5.) If $\forall w [w. \phi(y) \in \phi(LS)] \rightarrow \Box[\phi[y]\psi \rightarrow \psi_y]$ in all w

This means that if the LS update $\phi(y)$ belongs to the LS system of all worlds, then the LS update $\phi[y]\psi$ yields a new LS state with the property y in all worlds necessarily. A ubiquitous occurrence of a property across all worlds, constrained at a specific space-time event that is rigidly designated across all worlds, is identical to necessity in my framework, and constitutes a metaphysical assumption on my part.

The appearance of the LS rule in language becomes apparent if we map my formula unto natural language occurrences of analytic statements:

(4.1.6.) “All children are necessarily young.”

(4.1.6.1.) $v(\text{All children are necessarily young}) = 1 \leftrightarrow [\phi(LS)[\phi(\text{young}_1)]\text{children}] \rightarrow \text{children}_1[\forall x \exists y [W_x \wedge \text{Event}(\text{Time})_y]]$

(4.1.7.) “All bachelors are necessarily unmarried.”

(4.1.7.1.) $v(\text{All bachelors are necessarily unmarried men}) = 1 \leftrightarrow [\varphi(\text{LS})[\varphi(\text{unmarried-man}_1)]\text{bachelors}] \rightarrow \text{bachelors}_1[\forall x \exists y[W_x \wedge \text{Event}(\text{Time})_y]]$

Implying that a new instantiation of $\varphi(\text{unmarried-man})$ was introduced into the LS, which outputs Bachelor at all worlds W_x at the specific time and event that $\varphi(\text{unmarried-man})$ occurred. Bachelor and the predicate “unmarried man” then become identical and thus necessary in all worlds moving forward, as the predicate is subsumed by the entity and no longer occurs independently in the formula. It should also be noted that empirical matching is not a current concern to us, as my object concerns a universal representation of necessity rather than what is empirically necessary across all worlds (whether such a thing exists is dubious, in which case my necessary beings would be limited to natural laws and space-time phenomena). This is how analytic truths are represented.

Given this view, the approach to possibility becomes simple, as the structural form of $v(\text{Sometimes})$ is represented as Figure 2:

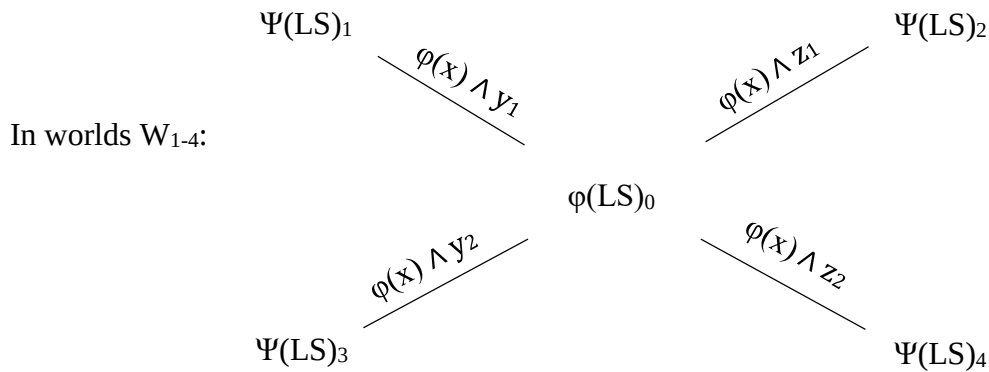


Figure 4: Structural representation of possibility in an LS system.

To represent how the property x is not ubiquitous as it was in necessity cases. Here, x occurs along some terminals at certain event-times (y_x) in some worlds while it occurs at other event-times (z_x) in other worlds. Therefore, emerging LS systems will contain mixed

emergent properties, such as y, z... or any other randomly co-occurring event-times in their worlds, making property x contingent.

An apparent and welcome consequence of this analysis is that alethic necessity becomes conflated with identity, similar to Leibniz's and Marcus' discussions. For this formula to work, one needs to accept that for the new $\Psi(\text{LS})$ to even contain a necessary entity, that entity needs to be one and the same with its properties. This tells us, for instance, that "three sides" is not an essential property of a "triangle", but one and the same as a triangle. Consequently, we are led to acknowledge that alethic modalities reflect tautologies in language that are analytic statements about the world, and that my conclusion seems closest to what Leibniz intended with his assertion that "in all true affirmative propositions, necessary or contingent, universal or singular, the notion of the predicate is always in some way included in that of the subject – the predicate is in the subject..." (Pickup, 2014, p. 5) yet in a sturdier formulation. It should be noted, however, that this equivalence between identity and necessity only applies to alethic modality, as we shall see with deontic necessity later. Additionally, the LS system seems to support Leibniz's law of identity between indiscernibles, as my system handles how entities come to analytically possess their properties in a necessary way, which intuitively does not support or imply that such properties may be entirely shared between different entities, as the LS system would simply recognize it as one and the same being. Lastly, it is striking that I have reflected Hegel's metaphysical scheme through the LS system, as one could argue that the notion of LSE is similar to what Hegel called sublation; that is, via logical-structural necessitation, an entity becomes necessarily identical with its set of properties as the two co-occur universally, and this necessary relation becomes "always the case" even retrospectively. While not truly identical to sublation, this might explain how things we have always known as "necessarily true" appear as such, for even if their LSE is later on from our current LS state, the very fact

that it is inevitable in the structure will cause it to spread across all points of reference – even to those where it hasn't been realized yet.

Despite using different methods, I have reached similar grounds to previous discussions on how necessity and identity are conflated; this method, however, is preferable as it grounds itself in ontology rather than metaphysical essences or mental states. Necessity has thus been defined while evading a reliance on essentialism and its associated vagueness and instead attributing what is often called “essences” to logico-progressive consequence.

4.2. Deontic Modality

The problem underlying deontic modalities are of a different philosophical ilk. We noticed how Von Wright (1951) and Kanger attempted to map formal logic unto the deontic forces of obligation and permissibility, yet fell short when it comes to providing a sufficiently explanatory account of what exactly constitutes deontic force in a sentence, and thereby in the world. Their definitions in 2.5.2.1. and 2.5.2.2. were descriptive and representational, but not explanatorily efficient on the ontological level of definition. My own account below prioritizes the clear identification of what characterizes an action as obligatory that functions efficiently in reference to either the past timeframe or the future. More importantly, Kratzer's (2012) work on ordering sources was similarly discussed. Kratzer attempts to define deonticity by making use of possible worlds semantics under the influence of an ordering source mechanism. If the modal base of a sentence, determined by the conversational background, merits a certain custom as normatively enforced, and if all the ordered (relevant) worlds accessible from this world also contain this custom as normative, then the statement itself contains deontic force. This was represented as the formula in 2.5.2.3. As noted in the review, I find reason to refute Kratzer's representations of deonticity due to both semantic and pragmatic issues. The reasons are as follows:

- a. A conversational background is a pragmatic phenomenon. If a modal base is formed via a contextually-determined set of n-tuples, then the members of this set cannot be quantitatively measured or reliably accounted for.
- b. The ordering source mechanism is sufficient for explaining how possible worlds contain identifiable worlds bearing explicit similarities to one another, but when designating deontic relationships, it lacks a stable indicator of what exactly is the threshold of accessible worlds that necessarily make a phenomenon “obligatory” in a local world. It suffers a similar quantification issue as point a.

I do not propose that that these two issues may be resolved while still maintaining Kratzer’s ordering source framework. My reasons for this are not limited to its impossibility, but rather to the view that obligation is not precisely quantifiable across possible worlds and does not need a modally global framework; obligation is variable across worlds infinitely, and even if identity is shared among them, none of these identities are rigid, as obligation is not a logical property in and of itself and may always be a subject to alterations originating from the world’s spontaneous conditions, unlike alethic modals. Finally, Belnap (1990) was also included in the review as a sort of precursor to what I’m about to propose. His approach essentially treated all expressions containing deontic force as imperatives rather than declaratives. While I do not do this exactly, we are about to see how deontics are, indeed, best defined as causal forces that function in a local framework rather than declaratives that are to be treated with the same global scope that “statements of fact” might be. As such, as an alternative view to obligation, I propose that a consideration of causality must be integrated into the analysis of deonticity, as it is the true driver of this phenomenon. I thereby move to my own framework on how to handle the notion of obligation in language.

I propose the use of new operators that operate on a causal-temporal scale in my already-established LS system, which I shall name the Causal-Temporal Projection

mechanism (CTP). This mechanism consists of two distinct operators: (PROACTIVE) and (RETROACTIVE), which I will abbreviate as (PRO) and (RET) in my analyses. It is crucial to understand how these operators work to understand any consequent formulation.

- a. When I use (PROACTIVE), we quantify ourselves by limiting our scope to two specific points in time, $\int_{t_0}^{t_1} \dots$, wherein the domain of (PROACTIVE) contains a Subject (S) who performs a specific act, and the point of time (t_0) at which the act is performed. The operator itself functions causal-temporally in a given world. Thus, within its assigned range, it checks whether the act (a) performed by (S) at (t_0) produces a specific effect (E) at a future point of time (t_1). If this effect is indeed instantiated, or is otherwise impeded by an empirical circumstance, the (PROACTIVE) operator is valid. This accounts for the sense of future obligation (must), as we perform an act for the sake of necessitating a future effect; whether or not this future effect happens or not is empirically realistic, accounted in the operator, and does not invalidate the obligation, by our use of $\emptyset E$.
- b. When I use (RETROACTIVE), we quantify ourselves by limiting our scope to two specific points in time, $\int_{t_1}^{t_2} \dots$, wherein the domain of (RETROACTIVE) contains a Subject (S) who performs a specific act, and the point of time (t_2) at which the act is performed. Within its assigned range, it checks whether an act (a) was ontologically necessitated at (t_1) in prospective consideration of (t_2). What this means is that the act at (t_1) was done for the purpose of producing an event at (t_2), much like how the (PRO) operator functions, and the operator (RET) checks whether this was the case. Of course, the event may have not taken place due to an obstruction, which may happen empirically, in which case we say $\emptyset E$. In this way, the (RET) operator verifies that (PRO) took place, hence accounting for instances of past obligation (had to).

One can take the following as the formalizations of the operators:

(4.2.1) $(PRO)(S, a, t_0, t_1, E)$ is true in a specific world (w) iff $a(S, t_0)$ causally projects to $E(t_1)$
or is blocked ($\emptyset E$)

(4.2.2) $(RET)(S, a, t_1, t_2, E)$ is true in a specific world (w) iff $a(S, t_1)$ was necessary for $E \leq t_2$
or is blocked ($\emptyset E$)

Furthermore, the structural forms of both operators may be represented as the following diagrams operating across time:

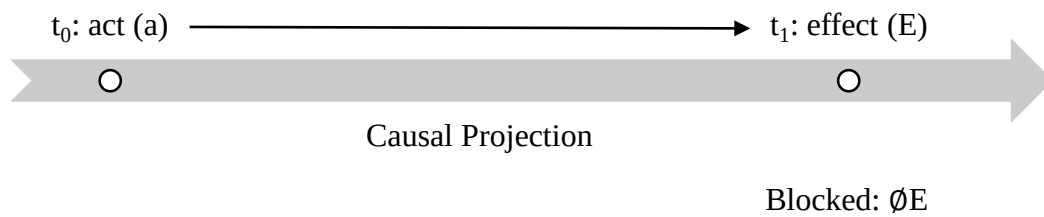


Figure 5: Structural representation of the operator (*PROACTIVE*)

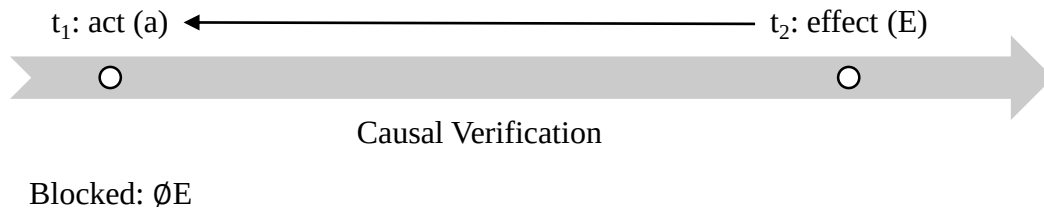


Figure 6: Structural representation of the operator (*RETROACTIVE*)

The formulae for past and future obligation may thus be represented:

$$(4.2.3.) \quad v(\text{Must}) = 1 \text{ iff } \int_{t_0}^{t_1} (PRO)_{S, t_0} [a(S, t_0) \rightarrow [[a(S, t_1)E] \vee \emptyset E]]$$

$$(4.2.4.) \quad v(\text{Had to}) = 1 \text{ iff } \int_{t_1}^{t_2} (RET)_{S, t_2} [a(S, t_1) \rightarrow [[a(S)E \leq t_2] \vee \emptyset E]]$$

In natural language, (4.2.3) operates over and checks a world-state at a future time (t_1) in accordance to if (E_{t_1}) is causally bound to (a) at (t_0). If (a) and (E) are causally bound between

both times, then $v(\text{Must})$ is validated whether or not E occurs in the future empirically; this accounts for how the subject “believes” they must act, but in ontological rather than psychological terms. (4.2.4) operates over and checks a world-state at a past time (t_1) and verifies whether the act at that time was necessary for the causation of an effect (E) at any time before (t_2), which is the moment of the proposition’s evaluation occurring. Once again, if (a) and (E) are causally bound between both times, then $v(\text{Had to})$ remains a truth whether or not E was obstructed empirically.

Finally, to map these constructions onto natural language uses, we may observe the following examples:

(4.2.5) I must study the exam to succeed.

(4.2.5.1) $\int_{t_0}^{t_1} (\text{PRO})_{I,t_0} [\text{study}(I, t_0) \rightarrow [[\text{study}(I, t_1)\text{succeed}] \vee \emptyset\text{succeed}]]$

(4.2.6) John had to go to the store to get groceries.

(4.2.6.1) $\int_{t_1}^{t_2} (\text{RET})_{\text{John},t_2} [\text{go_to_the_store}(\text{John}, t_1) \rightarrow [[\text{go_to_the_store}(\text{John})\text{get-groceries} \leq t_2] \vee \emptyset\text{get-groceries}]]$

These are mapped onto these language uses as they always represent the given structures of the world whether or not the reasons for the actions are explicitly stated (ex. “I must study the exam.”), for as I have discussed before, pragmatic constraints come after the world structure and are implied as understood; in my analysis, it is the reflected world structure that I am representing.

With these formulations, I am essentially accounting for the phenomenon of obligation in terms of ontological relations and empirical grounding, while also avoiding cognitive elements in the system. It is worth noting that this framework largely commits itself to the Block-Time theory of philosophy, as it assumes that past and future are existent entities

in much the same way as the present, which is not refuted by modern physics, but remains an open possibility. With an understanding of obligation covered, I now turn to understanding boulomaic modals.

4.3. Boulomaic Modality

Even more ungraspable by formal logic has been the notion of desire as manifested in language. For such a phenomenon to have a worldly base, it must be represented and explained in a stable manner much like the previous and following sections are done. As we saw with Hintikka (1962), informal systems may indeed be mapped to formal systems, but not without a great risk of metaphysical unclarity. Hintikka assigned belief-states in language a truth value depending on how compatible the given belief is with a set of accessible worlds from the ground world. His delineation of beliefs, which consequently translates into wish-states, is unfavorable to us, however, due to the following:

- a. Even though Hintikka (1962) states that the actual world w need not be a part of the assessed set of worlds for a subject's belief, such as in the case of absurd beliefs, and nor did he assert the existence of impossible worlds, a natural consequence of his framework is that illogical beliefs will entail possible worlds with illogical or impossible entities. This sort of ontology collapses the reality of axioms in any formal system. If a subject x believes " $\emptyset\phi$ " where ϕ is a logical axiom (ex. " p implies p "), then an inevitable outcome is logical explosion (exemplified by the principle of explosion) across possible worlds: A contradiction being true implies anything can exist. If such worlds exist, they de-fabricate the very notion of possible worlds semantics, as $\emptyset(p \rightarrow p)$ will effectively erase the world in which it holds true. Alternatively, if we avoid this while still maintaining Hintikka's system, it wouldn't be sound to claim that "no world set exists if x believes something contradictory.", as the belief itself still exists, and may be firmly believed indeed, while still remaining unaccounted for in the ontology. Others have noted this issue as well, and

proposed the existence of impossible worlds as logico-semantic realities to make this framework hold, but I view this as unnecessary if it is the only approach left for a theory to survive.

- b. Hintikka (1962) assumes that the subject x is rigidly identifiable across worlds where x holds the assessed belief y . This may be true in some cases, while failing in others, as the occurrence of $(X \wedge Y_x) \wedge (X_{w0} = X_{w1})$ is not guaranteed in all worlds where x believes y . A world where x believes y may hold, but x may be defined differently in that world; conversely, x may be rigidly identifiable across 2 worlds, but the belief y may not be rigid and lost to its original interpretation. This framework thus seemingly struggles with identification issues.

Regardless, my issue at hand concerns the notion of desire and not belief specifically, but this was an important foundation for attitude talk in semantics, despite the given critiques, as beliefs are categorically attitude modes much like desires.

I also went through Portner and Rubinstein's (2020) representation of desire. For them, the application of an ordering source mechanism upon desires is quite applicable; a propositional desire is evaluated according to a set of worlds that are ranked as "desirable" by the subject via what is called human necessity and a background of doxastic beliefs (much like Kratzer's conversational background). This was given as 2.5.3.1 in the review of literature. I noted how this is an impressive formulation, albeit it lacks a precise definition of what constitutes desire in a subject and how it, indeed, relates to an act that may or may not occur. A core argument of my thesis that is now common to both sections 2 and 3 is that wishes, just like obligation, do not need an ordering source framework explicitly rooted in possible worlds and such an approach only blurs their meanings further. A possible worlds framework with ordering sources is better-used in grander, philosophically complex notions

such as universal necessitation, impossibility, and, of course, possibility, while debating whether each tackled concept exists locally or globally across worlds. However:

- a. Obligations and desires do not present the need for global quantification, as the matter involves a human agent, who loses rigidity of identity the more we abstract into possible worlds.
- b. Wishes are not quantifiable in the way event-times are, as they lack stable domains that anchor them to their worlds for identification, resulting in the very same issue that its bearers possess.
- c. In treating wishes as a function of human necessity, which operates under the same rules as a typical ordering source, wishes are essentially reduced to another form of “I must act” due to x and y accessibility relations, based on favorability instead of normative rules, which both blurs the line between obligation and desire and is an even greater pragmatic-cognitive factor that we wish to avoid.

Finally, I considered the influence of DRT - primarily MSDRT, on how states of desire and belief may be evaluated. It certainly achieves the task of mapping a subject S's mental representations of reality unto the attitudes held by S, as exemplified by Figure 2. The extent to which MSDRT can be used to represent mental state evolutions is impressive, and indeed provides a solid formulaic system that makes sense of the link between our mental representations of entities and how our subjectivity interacts with them in a discourse. However, an issue apparent in this framework is the clear treatment of boulomaic states as primitive predicates upon entities that go more or less undefined in Kamp's system and immediately function of attributives. This is a similar issue shared with Portner's ideas, as we saw before. A semantic system is all the more fine-grained the more we decompose its “primitive” concepts into their semantic structures and integrate them as such into larger complexes in order to avoid philosophical vagueness, which will inevitably sully linguistic

clarity. This will be my goal in the framework beyond its ontological grounding. Moreover, DRT/MSDRT liberally makes use of natural language terms to account for preliminaries such as prepositions and semantic hypernyms, which are either too broad, undefined, or simply unnecessary in the structures. Lastly, of course, MSDRT is completely psychologistic and hence inherently opposed to my own framework and commitments. It superficially grounds boulomaic states within the mind and does not consider its logical status or function in the world, which comprises my main inquiry. Thus, Kamp's system neglects providing a fine-tuned, comprehensive semantic system of meaning and narrows down on discourse evolution or mental states rather than first addressing the status of what the linguistic meaning of "believe" and "desire" might be in a structural manner. On the basis of this, I now return to the claim of why a local assessment rooted in time and causation is once again appropriate, within which I will also introduce a new domain of definition pertaining to emotive potential in the world before these domains induce cognitive states in an agent.

As a start, let us make clear that any analysis we have heretofore attempted has been grounded in world-states, and that even the emotive power of boulomaics needs to be accounted for in suchlike manner. It is true that some, such as cognitivists and cognitive linguists, may oppose my view given the very nature of boulomaic modals as indicators of emotional mental states rather than anything happening in the exterior world, but this is where my distinction lies: My treatment, just as it has always endeavored, provides a pre-cognitive background of modality without denouncing the influence of the mind in its interpretations; as such, the models created by cognitive linguists may still hold simultaneously to my own, with the simple acknowledgement that my ontological model precedes the mentalist one. I shall discuss what this implies further in my discussion towards the end of this thesis. For now, let us consider the following metaphysical statement: There is an extricable link between the world of corporeal entities and the world of consciousness,

evidenced clearly by our very existence. There is, then, an implicit commitment to Cartesian duality in my framework, implying that it also contains the same problems faced in it: The exact nature of the link between body and mind. I do not claim to provide a proper philosophical answer to this, yet I do commit myself to the existence of what I call “emotive potentiality” in the world, modelled by:

$$(4.3.1) [S(x) \wedge E_n(\varphi)] \models \Diamond[[as(z) \leftrightarrow v(|E_n(\varphi)| \geq 1) = 1] \vee OP_E: \Diamond(Event-time(z')_{\Delta\delta + n})]]$$

This reads as: For the conjunction of any agent and a particular emotive potential in an object, it is entailed that an action or future-determined event-time may or may not occur in that world *bound* to this potential, given that at least one such potential exists. I must note that the use of a possible worlds operator here is not in the standard Kripkean model, but rather a weak version of local possibility; hence, “ $\Diamond$ ” in 3.1 reads as “It is possible for (a) to occur only in the given model of the base world, and not across possible worlds.” In other words, $\Diamond x$ is true as long as it is simply not excluded from the possibilities allowed in D_E – it may hold or merely fail to occur given the world’s circumstances. The formula itself will become more intuitive once it integrate it into the complete truth-condition of $v(want/desire/wish\dots)$. However, the reasoning behind the structure is that all desires are either direct motives for immediate action within the agent’s and outcome’s timeframe or are distant wishes for something in the future to occur. Moreover, an emotive potential is what we may essentially reduce to and designate as the causal link between the subject and a particular world-state that is tied to the subject’s desires, albeit this world-state may never come to fruition. One may thus model the truth condition for boulomaic modality as such:

Let:

- a. D_E denote the domain of emotive potential defined as: An ontologically bound domain around particular objects that allows for potential object-consciousness interactions within it.
- b. OP denote the set of all properties possessed by an object Q
- c. OP_E denote the set of properties in OP that possess emotive potential via D_E
- d. (a) denote an act done by the subject S that has causal strength in the world

Therefore:

$$(4.3.2) \exists x \exists y \exists z [[S(x) \wedge Q(y)] \in D_E] \wedge D_E: [(OP, OP_E, OP_E \subseteq OP) \in Q(y)] \rightarrow |OP_E| = n(\varphi) \equiv E_n(\varphi) \rightarrow \Delta\delta(x, y): E_n(\varphi) \rightarrow [S(x) \wedge E_n(\varphi)] \models \Diamond [[a_S(z) \leftrightarrow v(|E_n(\varphi)| \geq 1) = 1] \vee OP_E: \Diamond (Event-time(z')_{\Delta\delta + n})]]$$

This formula is the truth condition of the boulomaics $v(\text{want/desire/wish...})$ that reads as:

“For a subject S and object Q that are qualified to belong in the domain of emotive potentialities (by a consciousness-object metaphysical link), this domain will necessitate that Q contains a certain set of special properties OP_E that are sub-categorical in the D_E domain. This special set is limited in its number of members and is empirically equivalent to the set of “desirable” effects $E_n(\varphi)$ that may be produced by the object Q . Thus, within the time frame of both S and Q existing in the world, S ’s conjunction with the $E_n(\varphi)$ of Q entails that there exists a possibility for an action to occur by S that is casually bound to $E_n(\varphi)$, and this action may only occur if the value of $E_n(\varphi)$ is true, meaning that a possible desirable outcome derived from Q indeed exists. Alternatively, if S wishes for something that may not be realized via a direct act (ex. Wishing to meet someone by coincidence), then $E_n(\varphi)$ only implies that a certain event-time may or may not occur in the future that belongs to the agent’s wish-set in OP_E . A visual structure for 3.3 would appear as figure 7:

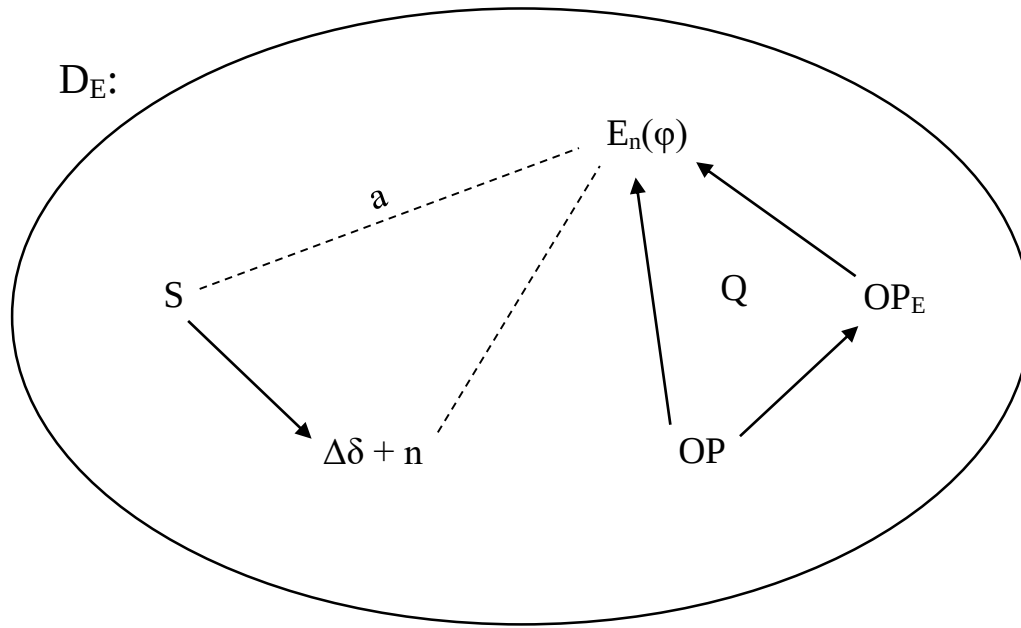


Figure 7: Structural representation of modal equivalents to “desire.”

We may see how the object Q is environed by its OP , OP_E , and the $E_n(\varphi)$ that is engendered by the set OP_E . $E_n(\varphi)$ is attempted to be reached by the subject S either directly via an act or via a process of “waiting” for a future actualization, both represented by segmented lines to show unreliability in their actualization. All of this takes place in the model of D_E , the domain that allows for emotive potentials. Applied to natural language, 3.3 would yield the following in the consequent direct and indirect interpretations of desire:

(4.3.3.) “Robert hopes to see Susan at the event tomorrow.”

For my framework to operate, it needs to be taken as implicit that Robert hopes to see Susan because of some particular reason, and that there is always such a reason even if it is not explicitly stated (manifested as $E_n(\varphi)$). Thus, the sentence is implicitly structured as “Robert hopes to see Susan at the event tomorrow (because x).” He may want to see her to discuss work plans, because he finds her beautiful, or any other reason that is “subjective” to Robert as it crucially belongs in the D_E domain. For this example, let’s assume Robert wants to see Susan because of one reason (OP_E is a singleton set): to discuss work plans, which I shall

abbreviate as [DWP]. [DWP] belongs to Susan's OP_E as the event-time of discussing work plans is causally anchored on Susan for Robert.

a. Interpretation 1: Robert goes to the event to purposefully see Susan.

(4.3.4.) $\exists x \exists y \exists z [[Robert(x) \wedge Susan(y)] \in D_E] \wedge D_E: [(OP, OP_{DWP}, OP_{DWP} \subseteq OP) \in Susan(y)]$
 $\rightarrow |OP_{DWP}| = 1(\varphi) \equiv [DWP] \rightarrow \Delta \delta (x.y): [DWP] \rightarrow [Robert(x) \wedge [DWP]] \models \Diamond[a_{Robert}(z) \leftrightarrow$
 $v(|DWP| \geq 1) = 1]$

b. Interpretation 2: Robert will be at the event regardless, but hopes to see Susan there.

(4.3.5.) $\exists x \exists y \exists z [[Robert(x) \wedge Susan(y)] \in D_E] \wedge D_E: [(OP, OP_{DWP}, OP_{DWP} \subseteq OP) \in Susan(y)]$
 $\rightarrow |OP_{DWP}| = 1(\varphi) \equiv [DWP] \rightarrow \Delta \delta (x.y): [DWP] \rightarrow [Robert(x) \wedge [DWP]] \models OP_{DWP}:$
 $\Diamond([DWP](z)_{\Delta \delta + n})]$

(4.3.4.) represents Robert's link to the desirable outcome $E_n(\varphi)$ via a direct action that it taken in an attempt to actualize it, while (4.3.5) represents a longing for $E_n(\varphi)$ that may take place at a future time.

This analysis of boulomaics raises an intriguing implication that indirectly reinforces a Lacanian approach to desire (wherein it is a quality in the domain of an object that is desired, rather than the object itself) completely unintentionally. It is only by $E_n(\varphi)$ existing that a subject may be linked to the object Q boulomaically and in turn "want something.". In turn, $E_n(\varphi)$ only exists, and may only exist, due to OP appearing in the D_E ; if an object lacks any emotive potential, then D_E will not comprise it. Essentially, we may take for granted that the presence of humans in a certain domain necessitates the D_E domain to appear in their mental and physical proximity, as there is no quantifiable limit to what we may want. This system provides an understanding of how desire links together a subject and an object of desire in a causal-temporal manner without diverging into possible worlds, makes use of a

new domain of definition that allows entities to interact with consciousness, and accounts for both interpretations of desire, direct and indirect, all while staying outside mental representations and not making use of psychological predicates. All of this occurs in the world before the subject is “aware” that he wants something; in fact, it is this causal structure that we built that allows the agent to realize, on the psychological level, that something is desired after it has been interpreted by the mind as it comprehends the boulomaic formula from the LS structure. This once again provides evidence for my thesis that modality is rooted in world-states before the mental stage.

4.4. Spatial Deixis

The primary issues underlying spatial deixis were noticed in the works of Kaplan (1979) and Evans (1991). Kaplan provided an interesting division between Content and Character, which account for extensional and contextual meanings respectively, as the latter maps contexts unto the former. The issue here is that in attempting to integrate contexts into semantic formulae, Kaplan essentially accounted for them via the parameters found in $\mathcal{U} = \langle C, W, U, P, T, I \rangle$, which are by no means sufficient indicators of context, as individuals, positions, times, and worlds are parameters all found in standard intensional semantics as well, while the interaction of C and I (contexts and function mappings), which seems analogous to his concept of character, simply maps the aforementioned four parameters to each other in formal form. How can context be unique in this formulation? If character operates with the same functionality and philosophy as possible world mappings (by mapping times and worlds to individuals and positions), then it too needs explicit accessibility rules and, more importantly, an explanatory metaphysics. I do not endeavor to provide such a thing for this theory, as I insist that the notion of character to content mapping is too derivative of the “possible worlds to extensions” mappings found in possible worlds and intensional semantics, which reduces Kaplan’s system into that of intensions and not the contextual

semantics for which he hoped. This system does not identify context as unique enough to significantly influence meaning, but rather as one element among various parameters that all eventually saturate into the concept of base intension. Lastly, as it clearly relates to my framework, Kaplan's account does little in providing a formal mechanism for deixis (or demonstratives) independent of context, as context is clearly taken for granted in his formulae. What I hope to establish is an account of deixis that is found rooted in the environment of the linguistic utterance – not contextually, but logically in the world.

Evans (1991), alternatively, discusses what he calls frames and cognitive maps. He provides a meta-epistemological perspective on spatial deictics by claiming that we refer to objects in space due to the passed down link of information from our previous knowledge of the entity to the present as it transposes into new space-time locations. As noted, this neatly, if not superficially, accounts for space, time, and causality as found in deictic expressions. However, it could stand to benefit from more formal representations of how causality exactly interacts with time and space, and in eventuality, a subject who is referring. I shall now attempt such a thing with my introduction of a new domain of assessment belonging to the LS.

We may begin with an account of the truth-condition of the spatial deixis “here” and build a bridge from it towards an account of “there”. As a first step to form my framework, let us situate my base parameters systematically and build from them accordingly.

Let:

- a. S stand for the subject under assessment
- b. S' stand for the domain of accessibility of S defined by the limit L(S):

$$(4.4.1.) L(S) = [L(W_S) \wedge [C(S) \text{ if } |E_C| = n \in W_S] \wedge P_L(S)]$$

$L(W)$ represents the physics-based limits of the world to which S belongs, $C(S)$ represents the causal reach of S if $E_C \in W_S$, implying that S may bring about a specific change of state (E) to any given object Q in a finite number of steps as long as this change occurs in the world of S , whereas $P_L(S)$ represents the perceptive field of the subject bound only by base eyesight (as given by the limit superscript). All together, we can see a particular scheme of what many may erroneously call “context” arise - a thing I name the domain of accessibility of S : S' . In this domain, there are three determiners of what constitutes the “here” of the subject S arranged by decreasing order in their degree of relevance and potential necessity in the domain. Firstly, in its loosest and barest form, the limit of the domain of S is constrained only by the laws of the Universe itself when no further causal or perceptual anchor is provided. In such cases, we may see uses of the word “here” to refer to “our world”, “this dimension”, “this plane of existence”, and other analogous concepts. Indeed, the only uses of the word “here” that may explicitly return a false truth-value is if the subject is referring to a world aside from ours; in all other cases found in W_S , “here” can be taken to be true at face value in virtue of $L(W_S)$, which is not much of a limit at all and seems to always apply analytically. Now, we move to a more specific domain: the extent to which S may affect what is found around them must be addressed. In some uses of “here”, the subject may be referring to “this city”, “this country”, “in this building”, and other cases where S has no direct visual input of the complete area. In this case, despite the subject’s visual field only encompassing a tiny segment of the referred-to area, they may still influence this area via actions that are quantifiable, or in other words, cause a change to occur in the domain by a limited number of acts. Lastly, in a highly specific domain, we see the use of “here” to refer to “this room” or “the exact place I’m found in”. These uses are characterized by the perceptive field of S , embodied not by their cognition or powers of interpretation, but by mere biological limits of vision (this would naturally entail that blind people use “here” using only the first two

parameters); hence, we call $x y z$ “potentially perceived” for items in the potential field of vision of S in their current position. It is safe to say that the vast majority of the uses of “here” possess truth-values aligned with causal reach and perceptive field, and often an interaction of both. Regardless, both of these parameters belong under the domain of $L(W_S)$, as they necessarily wouldn’t exist without it. I may thus add a clause to my original interpretation of $L(S)$:

$$(4.4.2.) L(S) = [L(W_S) \wedge [C(S) \text{ if } |E_C| = n \in W_S] \wedge P_L(S)]. \quad [[C(S) \leftrightarrow P_L(S)] \subseteq L(W_S)]$$

If this schema is followed, it becomes clear that $v(\text{here})$ has three alignments that determine its truth-value that may or may not exist simultaneously depending on speaker intentions, but this specification is one of pragmatic nature. So, “here” may constitute the universality of world-reference or the particularity of visually-determined areas. Combining $L(W_S)$, $C(S)$, and $P_L(S)$, one may represent the following structure of $L(S)$ with Figure 8.

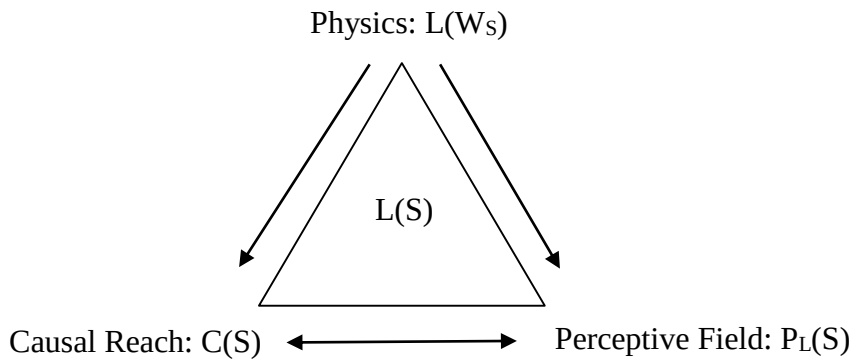


Figure 8: Structural form of the ontological limit of $v(\text{here})$

With $L(S)$ fully defined, we may proceed to integrate it into my system with decoding the parameters it itself is comprised of – I shall simply name it as $L(S)$ for brevity, and S' as the domain formed by applying its rules. Before doing so, let the accessibility relation also be defined:

c. A as an accessibility relation between S and S' , with A_{phys} and A_{prox} to refer to direct, physical accessibility, which is a subset of $P_L(S)$, and general proximity as we saw with

perceptual, causal, and worldly constraints. The intuition behind this is that accessibility may be physical or non-physical, yet each are limited by some manner in the $L(S)$; they are accounted for separately in the formula for coherence by acting as umbrella terms allowed by the parameters of $L(S)$.

Let us then take a look at the following formula:

$$(4.4.3.) \ v(\text{here}) = 1 \text{ iff } (S \wedge S') \rightarrow W_S: \{A, S, t_s\}: [[S. A_{\text{phys}}(S')] \vee [S. A_{\text{prox}}(S')]] \wedge d(S, S') \leq L(S)$$

The truth condition of “here” reads as: If a subject refers to their deictic domain S' , then this domain contains at least one accessibility relation alongside the subject at a certain time that is either a relation of physical proximity or non-physical (proximal) proximity, as long as the ontological “distance” between the subject and the referred-to domain is found within the bounds of $L(S)$. The structural form of $v(\text{here})$ may be presented in such a manner as figure 9. This shows how in order for a S to refer to the field of S' , the ontological distance between both needs to be mediated by the parameters of $L(S)$ defined in 4.4.2., wherein the accessibility relation between S and S' is aligned with either physical accessibility such as touch or vision, or proximal accessibility such as finite causal reach as defined in $L(W_S)$. If this schema is true, then it would adequately account for what is typically referred to with the deictic “here” and effectively supply us with an intensional structure of it.

To address the truth-value of “there”, I will need to decompose the structure of $L(S)$ in my formula. Most importantly, there is no limit to what “there” may refer to, as it is unbound even by $L(W_S)$ and falls outside the structure of figures 8 and 9. One may use the word “there” to refer to a counter in front of them just barely out of reach, or to fantasy world they read of in a book. This is why I abandon the use of $L(S)$ as a complete system in this case and only make use of its parameters as needed.

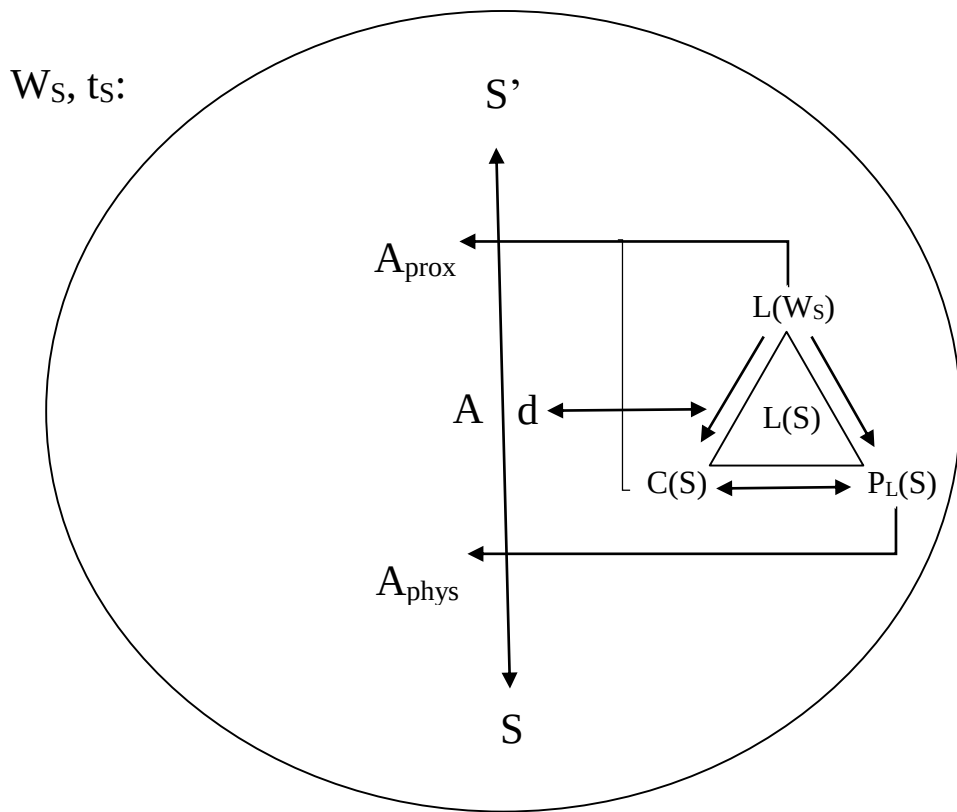


Figure 9: Structural representation of the truth-condition of $v(\text{here})$

Furthermore, when a subject refers deictically by “there”, they are doing the structural opposite of what they do when they use “here” – rather than referring to a domain found around them like an ontological bubble, they are rather excluding this very domain from their reference. So, S' still exists in $v(\text{there})$, but while it functions like an area in $v(\text{here})$, it now functions as a domain of exclusion that detaches itself from the ontological structure of “here”. To elaborate, “there” can only ever be a subset of “here”, as the latter exists universally while the former has a much more designative nature, yet at the same time, it may designate without limits. Some might argue that this designation may only be defined by a speaker, and while this is in eventuality true, one may also define it structurally by deriving it from $v(\text{here})$. This derivative method is truly the most and only precise way in which one may formally account for a word as deictically loose as “there”, which, if $r(S, S')$ stands for the relation between S and S' , might be represented as:

$$(4.4.4) \ v(\text{there}) = 1 \text{ iff } (S \wedge S') \rightarrow W_S: \{A, S, t_S\}: [[S. A_{\text{prox}}^+(S')] \wedge \sim [S. A_{\text{phys}}(S')]] \wedge r(S, S') \notin L(S)$$

Thus, S' is a domain only proximally accessible to S (as defined in $L(W_S)$ by causal reach and perceptual field) but is nevertheless found outside of the holistic nature of $L(S)$. In this sense, “there” refers to something that spots a hole in the field of $L(S)$, while nonetheless still making use of some of its faculties of proximity and following its structural rules. The script “prox⁺” is used to represent that the reference of “there” may fall outside of what is normally allowed by $L(W_S)$, such as referring to other worlds or places of fantasy on top of what is found in $L(W_S)$. Thereby, one may take the semantic structure of “there” to be analogous to figure 10. I have not omitted the existence of the limit $L(S)$ from the definition, as it is present to represent the severed link between S and S' found in A . So, if the structure is followed, S deictically refers to S' using “there” if S' falls explicitly outside the complete field of $L(S)$ but may still refer to a specific field within $L(W_S)$, $C(S)$, or $P_L(S)$, with the exception of A_{phys} , and may even extend beyond the limits of $L(W_S)$.

From the analysis itself, we may see that context does not need to be defined in any way for deixis to be represented formally, as it is a pragmatic, informal notion determined by speaker backgrounds and intentions. None of these are necessary qualities to understand what deictic function “here” serves, as I have managed to define “context” semantically and inadvertently reduced it to yet another causal-ontological phenomenon that does not truly merit the word “context”. This system also allows us to scale deictic reference by granularity, with “here” being the most “primitive” reference that may refer to the world, an abstract field, or a concrete one in decreasing gradation; other indexes will simply act as derivatives of this – an area worthy of further analysis.

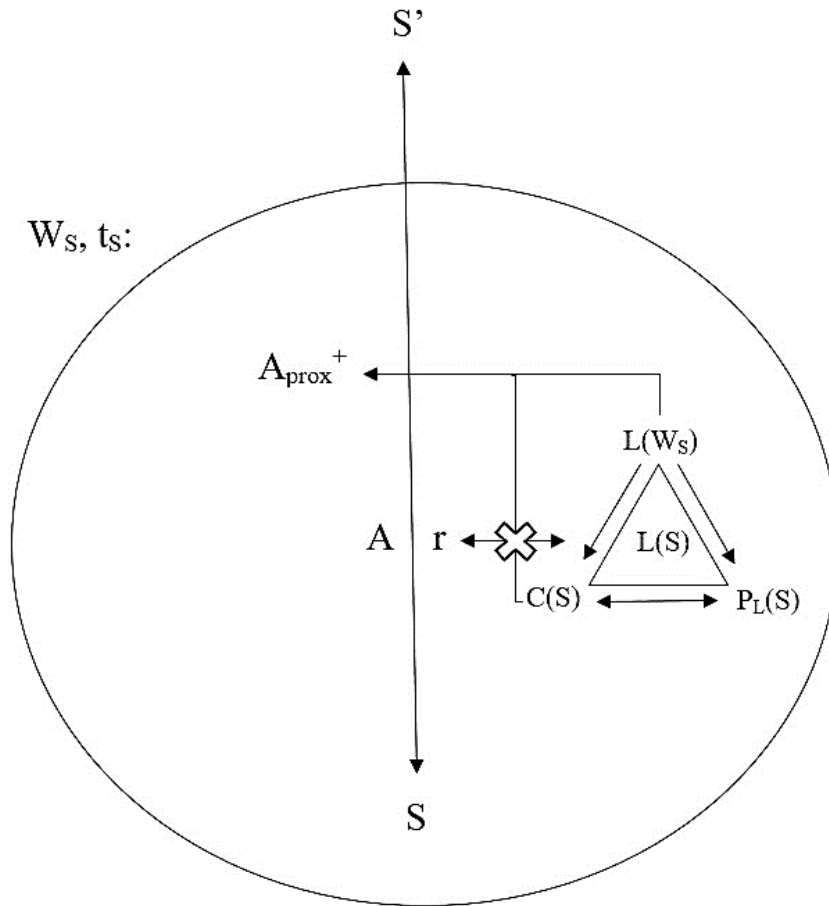


Figure 10: Structural representation of the truth-condition of $v(\text{there})$

Even more strikingly, this mechanism is what creates linguistic vagueness in deictic propositions, as speakers refer to the very structure I just formed by uttering “here”, “there” and beyond. What these words refer to cannot be understood unless an individual is aware of the ontologically, causally, and perceptually bound field around the speaker and integrates it into their understanding of the spoken units, which is the naturally proceeding cognitive step. This is why it is crucial to first have an understanding of what this structure is before introducing it to cognitive linguistics.

4.5. Temporal Deixis

As the final modal type I aim to address, temporal deixis was investigated in the work of Reichenbach (2005). Unlike previous sections, this section will centralize the work of

Reichenbach as an acceptable framework that needs further formal stability for the parameters of reference time and event time – one that I aim to provide. Although Reichenbach primarily uses these notions to account for past and future tenses, I derivatively make use of the notion of reference time, speech time, and event time in my analyses of simultaneous deixis (“now”) and the more temporally loose index “then” as elements in my own framework in parallel to others. One crucial note, however, is that I will not be integrating the idea of speech time in my framework, as I believe it to be a pragmatic constraint and one that is irrelevant to my typically retrospective analyses, since I aim to address what in the world itself necessitates the use of temporal deixis, so an utterance has not yet taken place in the system. Furthermore, I do believe both event and reference points may be more formally defined if one is to integrate them into formal semantics, and that is an undertaking I shall attempt. Thus, what will follow will be my formalization of Reichenbach’s two temporal reference points, followed by an analysis of $v(\text{now})$ and $v(\text{then})$ via their use.

Initially, let us consider what it means to be a “reference time”. For one to refer to a time, there needs to be an anchoring point from which the reference happens; as such, one must first situate an agent located in such a moment that may be called the “anchor time”. An anchor time may be analogous to my previous concept of the (PROACTIVE) and (RETROACTIVE) operators in that it assesses a specific point of time and its encompassing event(s) in accordance to a past or future point of time. However, it can be noticed that an anchor time is more flexible than those operators. It is assumed that the anchor time is a specific point (t_0) that I will refer to as (t_s) to mean “the point of time occupied by the subject”. It is worth noting a common attribute in both this and spatial deixis: that the self-reference point, often disguised as the personal deixis “I”, never changes, as “I” always refers to the same extension in any world. This merits further discussion, yet the current scope does

not encompass personal deixis – a topic deserving its own paper. As of now, the above can be taken to be an axiom for both deictic types and what holds the subject S in a stable spatio-temporal location. With this settled, one may formally define t_S as:

$$(4.5.1) \ t_S = t_0 \in W_0 \wedge E_{t_0}: \{E_1, E_2, \dots E_n\}$$

This implies that the subject's own location in time is identical to the “current” point of time t_0 , which belongs to the current world and contains a set of events that may be occurring at this point. Then, with t_S defined, I move back to the notion of reference time as it relates to t_S . Unlike spatial indexes, temporal ones cannot be defined via fields or limits, as they refer precisely to a definite point of time at which a definite event occurs. Thus, the LS system accounts for this via causal anchoring. A referred point of time ought to have at least one causal influence upon the anchor time, without which reference would be impossible for the subject as it would fall under a different domain. The empirical notion behind this is that a subject would have no knowledge of a certain point in time if they were not aware of at least one link between that point and their own. One may illustrate this as such with t_R standing for reference time and C for the set of potential causal sources in W_0 :

$$(4.5.2) \ t_R := t_x \in C: \{C_1, C_2, \dots C_n\} \wedge [t_x \notin C \rightarrow t_x \notin t_R]$$

So, t_R cannot occur if there is not at least one causal link between the reference time and the anchor time. With this in mind, the event time (t_E) may be brought into the equation by realizing it occurs within t_R as Reichenbach had stated. With this taken as true, then t_E can be defined as:

$$(4.5.3) \ t_E := (t_E \subseteq t_R) \wedge (C_E \subseteq t_R)$$

Meaning that both the causes and the events in the referred-to point of time are elements in the designated reference point. This leads one to consider whether the reference time is really an interval rather than a point, and this is a tricky issue, as no true “point” of time may exist

in the sense found in physics, but rather infinitely stretched intervals that are considered as points; without going into such details, all my “points” in time may be considered to be infinitely open to any number of events and causes, which is how t_E can belong in t_R .

Furthermore, when speaking of causes, it bears mentioning that the “event” spoken of at t_E is the very causal link that was defined in 4.5.2., as subjects refer to these points due to their influence on their own time point, which is precisely what C_x is:

$$(4.5.4) (t_R \wedge t_E) \in C: \{C_1, C_2, \dots C_n\} \wedge C_x \equiv t_E$$

With all these parameters defined, I now move to the complete formulations beginning with the truth-condition of $v(\text{now})$.

To follow up with the three time points discussed, the reference of “now” appears to present an analogous mapping among t_S , t_R , and t_E . Indeed, all three points occur simultaneously in this scheme and are equivalent in their identities. When a subject S uses “now” or its equivalents (“at this moment”, “right now”, “at present” ...) as a reference point, then they are in fact referring to their own anchor point, and so the reference time and anchor time become indiscernible ($t_R \equiv t_S$), while t_E refers to an event in t_R as usual, which also now belongs to t_S :

$$(4.5.5) v(\text{now}) = 1 \text{ iff } \exists x (|C_x| \geq 1) \in (t_R \wedge t_S) \rightarrow (t_R \equiv t_S) \wedge t_E \subseteq (t_R \wedge t_S)$$

This tells us that if at least one specific causal link exists that belongs to both the reference point and the anchor point, in that S will end up referring to an event(s) that co-occurs in their anchor time and the intended reference time. This implies that the reference and anchor points are indiscernible or possess the same extension, while the referred-to event E belongs under both of these domains (or rather, this combined domain). One may illustrate this with figure 11. This figure shows us how t_S and t_R occupy the same referential domain, within which we have a causal event linked to the subject.

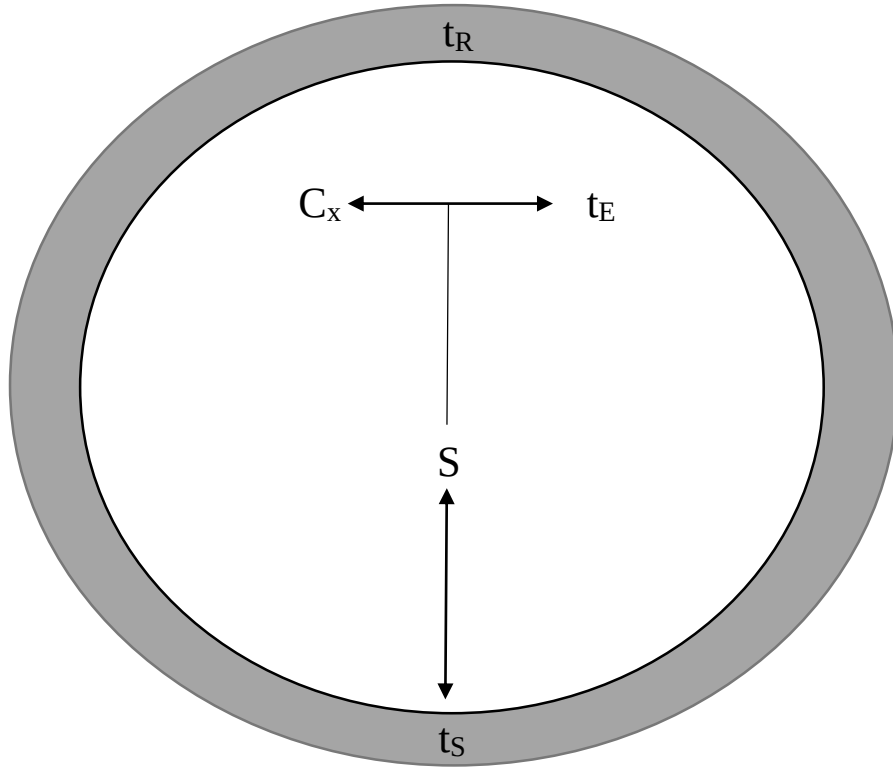


Figure 11: Structural representation of the truth-condition of $v(\text{now})$

It is a simple illustration to represent what is in eventuality a simple notion. The matter gets rather tricky, however, once we consider what this system implies for $v(\text{then})$, as the reference of “now” is once again taken to be a primitive when evaluating other references, as was done for the spatial deixis “here”. With this in mind, the vaguer notion of “then” may be considered, which is perhaps even ontologically looser than $v(\text{there})$.

As is consistent with my above structures, the extension of “then” is taken to refer to any reference time not found in t_S itself, and that this is the extent to which $v(\text{then})$ may be formalized on the pre-contextual level. Crucially, when we exclude t_S from the structure of $v(\text{then})$, we are essentially omitting the present interval from the interpretation and claiming that t_R may only exist in either a past or future moment, and never both. One may take this to symbolize:

$$(4.5.6) \ v(\text{then}) = 1 \text{ iff } \exists x (|C_x| \geq 1) \in [t_R := (t_{S-n} \oplus t_{S+n}) \wedge \sim t_S] \wedge t_E \subseteq t_R$$

This can be read as: There exists at least one causal link in the reference time that falls either before or after the anchor time and never inside, while the referred-to event falls under the reference time. This can be illustrated by figure 12:

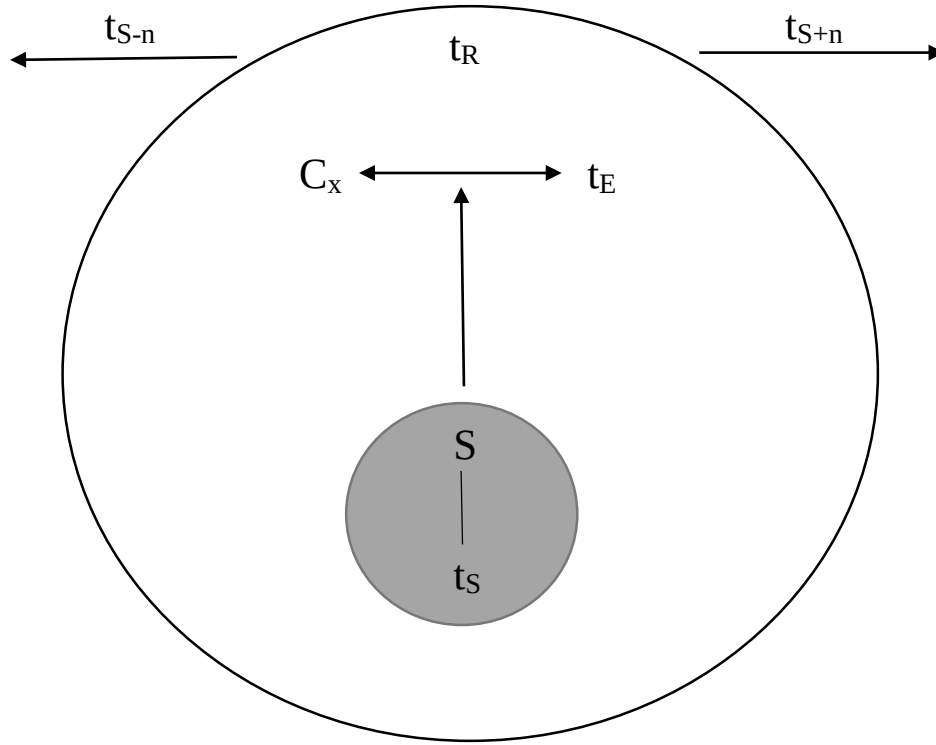


Figure 12: Structural representation of the truth-condition of $v(\text{then})$

We might see how t_S is now situated in a limited domain only pertaining to the subject S and bears no direct relation to t_R besides its association by causal reference. Then, as expected, the subject refers to a certain causal event under the domain of t_R that explicitly falls outside the anchor time. The strength of this interpretation is that “then” can refer freely to any reference time whether it is situated in the past or the future, as seen in natural language use. Empirically speaking, one may use “then” as “Meet me then by 10 o’clock” or “I remember what you told me then”, which refer to a future reference point and past point respectively. By allowing t_R to simply stand outside the domain of t_S (the present interval), we necessarily allow t_E to be located in any point of time that is not the present.

The flow from spatial to temporal deixis has been smooth in this way, as both take the subject's spatio-temporal environment to be a primitive from which other reference points are created by the subject. Much like my CTP system for deontics, this once again relies on the Block theory of time to allow causal links to hold from one time point to the other in the case of temporally proximal indexes (ex. "then"), albeit not to the same degree as CTP; this is not a reliance I view in ill regard, because the Block-time theory has solid foundations in both the philosophical and scientific literature, and some of my theories' reliance on it can be viewed as a fortunate coincidence, or, if my theories are logically sound, perhaps even an evidence for it.

Thus, Reichenbach's original theory of temporal deixis has been reformulated by formalizing his parameters, adding my own, and combining them to suit the LS framework. Although I have not taken multiple perspectives as my previous sections, I believe Reichenbach's distinctions for types of references are already in line with my formal commitments, if not for a want of some sharpening. That is precisely what was done, in that Reichenbach's parameters of temporal indexes were run through a filter that locates itself in pre-contextual ontology, emerging with a system that is analogous to it yet much more definitive in what those parameters refer to and their consequent truth-conditions.

4.6. Answering the Research Questions

With the analyses completed, let us now look back at the original research questions with which I began this thesis and provide suitable answers to them as per my investigations.

4.6.1. Answering the First Question

To what extent can we ascertain the definitive truth-conditions of modal and deictic propositions using purely logical systems of representation?

It appears clear to us, as my formulae suggest, that we may indeed utilize model-theoretic semantics to account for linguistically “vague” categories such as modals and indexes. I have provided not just definitive truth-values for each category, but also the truth-conditions under which they are valid, all the while upholding the formalism of Frege, Tarski, Montague, and their peers. Hence, it is safe to claim that we may to a great extent, and if my work is further expanded – fully, ascertain the truth-conditions of modal verbs and deixis as they are used in natural language by linking them back towards the world-states they are derived from.

4.6.2. Answering the Second Question

What are the necessary ontological commitments that must be addressed before attempting formal representations of modality and deixis?

This was of central issue in this thesis, and as was seen, in almost all other scholarly work in semantics. My own ontology supports that of complete formalism as it is rooted in an assumed world structure. This structure, indeed, is the primary ontological commitment of this particular thesis. However, as was encountered in my analyses, a great number of semantic studies do not consider or address their own commitments before providing their frameworks, such as Barcan Marcus’ commitment to the existence of domain-bound necessitation, Kratzer’s commitment to the existence of ordered worlds applying to obligation states, Kamp’s commitment to belief and desire as primitives, and Kaplan’s commitment to context as a primitive. These are all issues I addressed and avoided in my own formulations, as they were left unaddressed in their original corpuses and thus weakened the truth-conditional stability of the frameworks. As such, the ontological commitments that must be addressed before attempting formal representations of modality and deixis specifically are those that either psychologically or pragmatically contaminate the formalism of the analyses

via commitments to unfounded primitives or, in some cases, metaphysical systems that mismatch with or misrepresent the analyzed linguistic types.

4.6.3. Answering the Third Question

How do the theories of other linguists and philosophers hold up against the rigid objective commitments of formal semantics?

As evidenced from this chapter, the majority of theories in formal semantics hold great merit, yet they suffer from philosophical vagueness. The contemporary frameworks in the field, albeit not all of them, noticeably diverge from the original philosophy of formal semantics set down by the likes Frege and Tarski, in that the very goal of systematic analyses in this field is to root itself in objective primitives (axioms) and world-states rather than rely on cognition and pragmatics as facilitators. Certainly, some modern researchers might have issue with this rigid definition of formal semantics, but I believe this was the original intent of its founding figures: An analysis of meaning via methods not reliant on natural language as an analytic tool, and to do so requires the exclusion of its associated ambiguous properties rooted in the mental world and the contextual world. In this sense, a surprisingly large number of researchers do not comply with this commitment and resort to the use of variables and primitives that are the domain of cognitive linguistics and formal pragmatics, but not formal semantics proper.

Chapter 5

Conclusion

What may be reliably drawn from this study is that not all linguistic features in language are cognitively rooted, and that the number of linguistic features derived from the world itself is likely much greater than originally conceived. Furthermore, the triangular relation found within the world, language, and the mind, is not an omnidirectional one; rather, if this thesis holds merit, the world can be said to be the originator of linguistic meaning from which the mind derives abstract complexes such as the ones I have laid forward, upon which cognitive and pragmatic variables are imposed. Through this, language attains what we call meaning – through a combination of abstract world-structural formulae and cognitive-pragmatic additions, language reaches its “comprehensible” stage for both its user and receiver. What this thesis achieves is that it represents the former contributor to meaning in specific semantic environments, the much less analyzed half in the literature compared to the latter half often analyzed by cognitive scientists and pragmatists.

5.1. Summary

On the course of the thesis, I followed a systematic pattern for all five analysis sections. I began with alethic modality and laid forward what classical thinkers assumed of the matter, soon coming to the realization that necessity is not strictly equivalent to metaphysical essences, but rather it constitutes identity itself. When we speak of necessity, it will not do to rely on indiscernible relations of properties among objects as a basis for making necessary identity statements for those objects, as properties and their holders may shift across worlds, along with the fact that indiscernibility does not simply “occur” in the world without a backgrounded world-structure to explain it. In response, I formulated such a

structure, albeit a rudimentary one, to explain how the world evolves in and of itself and comes to possess collections of properties at specific event-times to give rise to objects that are identical to their co-occurring properties. I ended up acting as a support to Leibniz's and Hegel's overall metaphysical philosophies by the formulation of the LS system, which also served as a base for the consequent analyses. Regarding deonticity, it was realized that they pose the troublesome issue of introducing human action upon what was previously a barren formal logical system. As scholars like von Wright (1951) attempted an analogous mapping, it was immediately clear even to them that there is a disparity between the logical formulas of necessitation and that of the much more specific obligation that only applies to acts. It was observed how Kanger (as cited in Hilpinen, 2001) and Kratzer (2012) provided respectable solutions to this problem, yet their theories lacked a proper explanatory nature of what makes an act obligatory in favor of descriptive theories and, in Kratzer's case, relied on the ontological assumption that an ordering source framework across possible worlds may extend towards the reality of obligation. As a more grounded solution, I proposed the CTP framework that relies purely on causality across time in a given world – a much more quantifiable notion. Through my new operators, I explained how obligation comes to be via a causal relation between time points that is found in the world's logical structure; this framework accounts for obligation as it is empirically seen, since it does not require the effect to truly happen for an act to possess an obligatory force in a given moment, but only on whether it is indeed tied to an act causally. For the final modal I tackled, boulomaics presented yet another issue of human agency within a formal system. Previous theories attempted to either treat beliefs and desires as semantic primitives, or tried to map an ordering source framework unto wishes, both of which do not sufficiently explain what necessitates a want as we know it in language. As a response, I rooted modal desire in a causal system that is found in a specialized domain that bridges the Cartesian halves of body-mind, through

which a set of an object's properties may transform into desirable effects that are causally linked and thus pursued by a subject under this domain. This may lead to one of three things to attain this effect: An immediate act, a wait through time, or a failure of an effect to transpire after either of the first two paths. Regardless of the outcome, the presence of this structure is what gives rise to a modal use of desire in language. When it comes to deixis, we noted its great degree of reliance on contextual cues out of which their meanings are derived that constitutes its semantic vagueness. To tackle spatial deixis, the theories of Kaplan (1979) and Evans (1991) were addressed, the former of which derives a content-character function from the archetypical intension-index function of most modal semantics; this makes Kaplan's insistence that context maps meaning to propositions via their character a mirror image of how an index (or possible worlds) maps meaning to intensions via their extensions in those indexes/worlds. While modal logic set its claims upon pre-laid, rigorous philosophies, Kaplan's system becomes an unphilosophical bastardization of those claims by treating context as a reliable ontological parameter. Evans, on the other hand, provides some favorable philosophical arguments that support my thesis, such as his proposition of frames of reference, spatial relations among objects, and egocentric spaces. These concepts are rather vaguely defined, yet they do correlate with the notions used in my framework, which is why they are mentioned. As for my own system, I proposed a new ontological relation among subjects and objects called "accessibility relation" that is backgrounded by three variables: Physics, causal reach, and perceptive field, that may be scaled from the most distant to the most immediate accessibility relations. According to the variable at play, a subject will establish a certain accessibility relation to their deictic domain that will be referred to by their deictic expression. Finally, for temporal deixis, Reichenbach (2005) was taken to be my theoretical pillar for my own system. I provided an overview of his notions of reference time, event time, and speech time, and, after filtering out speech time from the system, I adopted

the former two into a new system. The system thus proposed measures the movement of the reference time in relation to the subject's time by situating the event time in a causally relevant time interval under the reference time, which may overlap with the subject time or fall beyond it. All together, the five analytic sections provided formal mechanisms for three modal and two deictic categories with no reliance on unpredictable variables of context and the mind. A visual summary of my findings may be found on the last page of the conclusion, given in table form.

5.2. Limitations of the Study

The primary setback of this thesis is its narrow scope of linguistic categories. For the framework of LS to truly hold in any ontological or semantic system, it needs to be further corroborated by linguistic categories of all sorts, beyond simply those of three modal and two deictic types and verifying whether they hold in a logico-structural ontology as well. Furthermore, a secondary weakness of a more philosophical nature can be identified as the presupposition backgrounded all throughout the thesis – that it is, in fact, a metaphysically sound claim to make that the world's metaphysics allows for the LS realm to exist, in the same sense that Frege's logical realm faced this issue. While the LS realm has more constituents and machinery than Frege's more incomplete proposition, it still needs a dedicated philosophical exposition as to show why and how this realm exists, and how it interacts with the mental and physical realms to produce its proto-semantic systems.

5.3. Recommendations and Implications for Further Research

The propositions of this thesis open the way for multiple angles of linguistic and philosophical expansion. Linguistically, further semantic categories that influence meaning, mainly tenses and determiners, along with other categories of modals and deixis, may be studied for their degree of compliance with the LS system that I have established. If those categories hold in the world using similar formal methods to the ones so far used, and situate

themselves as a part of the LS system, they shall further stabilize the ontological claim made here about language. Moreover, it would be prudent to expand this framework into a syntax-semantic interface by connecting the formalizations of the LS system with how they may manifest themselves in different syntactic structures and possess altered meanings accordingly; this would be more of a supporting study, as syntax relates more to language rather than the world, but it would still corroborate the stability of my system if its relationship with word order and clause embeddings is investigated. Lastly for linguistics, one might chronologically proceed from the LS structure to its next natural step – pragmatic and cognitive transformations. It would provide my framework with great generalizability if the transformations of the logical formulae I have provided are investigated once they finally reach a human encoder. How do the formulas morph under pragmatic-cognitive constraints? What parameters and operators are added to them? These would make for fascinating studies to build upon this ontology. As for philosophical expansions, one might study the exact nature of “predicates”, “operators”, or “parameters” as they are found in ontological states. These studies would not be of a formal nature, but rather in a prosaic, classical manner more parallel to the “continental” tradition than to its analytic descendant. Lastly and certainly, as mentioned in the limitations, a proper exposition of a logical realm may be provided in a dedicated metaphysical discussion. These expansions would provide solid ground upon which the theory I defend may stand more comfortably and verifiably in both the linguistic and philosophical literature.

As a closing illustration, the following page lays out the results of each of the formal representations that were developed in this study as Table 1. Each formula is given as its final, post-developed version.

Truth-condition of alethic necessity under the linguistic usage $v(\text{always})$ and its equivalents.	$v(\text{Always}) = 1 \leftrightarrow [\varphi(\text{LS})[\varphi(y)]\psi] \rightarrow \psi[\forall x\exists y[W_x \wedge \text{Event}(\text{Time})_y]]$
Truth-condition of alethic possibility under the linguistic usage $v(\text{sometimes})$ and its equivalents.	$v(\text{Sometimes}) = 1 \leftrightarrow [\varphi(\text{LS})[\varphi(y)]\psi] \rightarrow \psi[\exists x\exists y[W_x \wedge \text{Event}(\text{Time})_y]]$
Truth-condition of future deontic obligation under the linguistic usage $v(\text{must})$ and its equivalents.	$v(\text{Must}) = 1 \text{ iff } \int_{t_0}^{t_1} (PRO)_{s,t_0} [a(S, t_0) \rightarrow [[a(S, t_1)E] \vee \emptyset E]]$
Truth-condition of past deontic obligation under the linguistic usage $v(\text{had to})$ and its equivalents.	$v(\text{Had to}) = 1 \text{ iff } \int_{t_1}^{t_2} (RET)_{s,t_2} [a(S, t_1) \rightarrow [[a(S)E \leq t_2] \vee \emptyset E]]$
Truth-condition of boulomaic desire under the linguistic usage $v(\text{want})$ and its equivalents.	$v(\text{want}) = 1 \text{ iff } \exists x\exists y\exists z[[S(x) \wedge Q(y)] \in D_E] \wedge D_E: [(OP, OP_E. OP_E \subseteq OP) \in Q(y)] \rightarrow OP_E = n(\varphi) \equiv E_n(\varphi) \rightarrow \Delta\delta (x.y): E_n(\varphi) \rightarrow [S(x) \wedge E_n(\varphi)] \models \Diamond [[a_S(z) \leftrightarrow v(E_n(\varphi) \geq 1) = 1] \vee OP_E: \Diamond (\text{Event-time}(z')_{\Delta\delta + n})]]$
Truth-condition of deictic anchored spatial location under the linguistic usage $v(\text{here})$ and its equivalents.	$v(\text{here}) = 1 \text{ iff } (S \wedge S') \rightarrow W_S: \{A, S, t_S\}: [[S. A_{\text{phys}}(S')] \vee [S. A_{\text{prox}}(S')]] \wedge d(S, S') \leq L(S)$

Truth-condition of deictic unanchored spatial location under the linguistic usage $v(\text{there})$ and its equivalents.	$v(\text{there}) = 1 \text{ iff } (S \wedge S') \rightarrow W_S: \{A, S, t_S\}: [[S. A_{\text{prox}}^+(S')] \wedge \sim [S. A_{\text{phys}}(S')]] \wedge r(S, S') \notin L(S)$
Truth-condition of deictic anchored temporal location under the linguistic usage $v(\text{now})$ and its equivalents.	$v(\text{now}) = 1 \text{ iff } \exists x (C_x \geq 1) \in (t_R \wedge t_S) \rightarrow (t_R \equiv t_S) \wedge t_E \subseteq (t_R \wedge t_S)$
Truth-condition of deictic unanchored temporal location under the linguistic usage $v(\text{then})$ and its equivalents.	$v(\text{then}) = 1 \text{ iff } \exists x (C_x \geq 1) \in [t_R := (t_{S-n} \oplus t_{S+n}) \wedge \sim t_S) \wedge t_E \subseteq t_R]$

Table 1: A table showing the results of the formal representations structured in this thesis

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